

Human Computer Interaction

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Lecture 06, Part 1: Evaluation techniques and sample systems

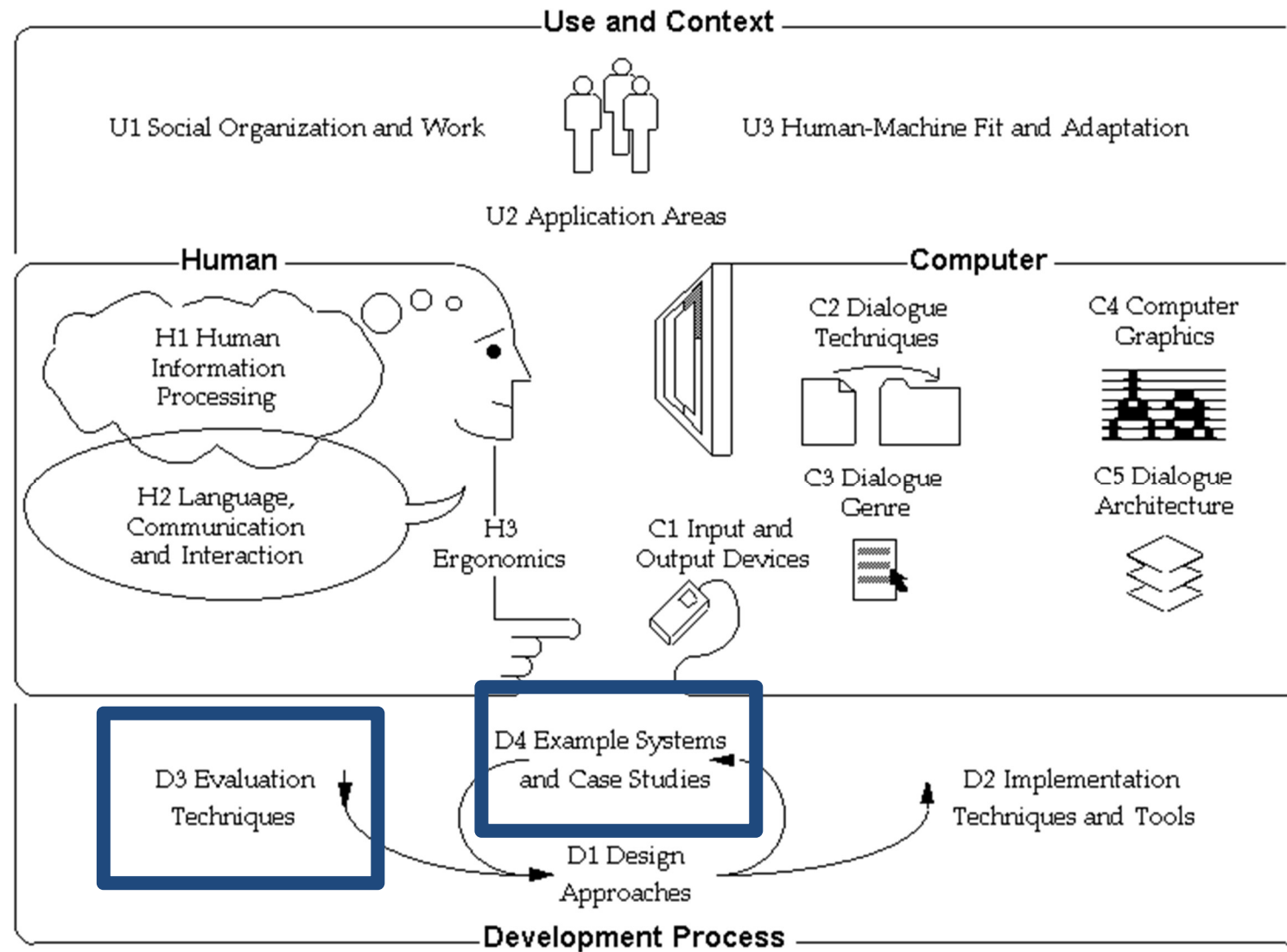


“I am programmed to understand humans”

C-3PO (“3PO” aka Threepio)

Lukas, G., & Hales, J. (2002). *Star Wars Episode II: Attack of the Clones*. Los Angeles, CA: 20th Century Fox. [Part 7/10] - 1080p HD, 7:00 min. Retrieved May 21, 2012 from http://www.youtube.com/watch?v=-o5Y_AszZGg

Overview



Contents

- ❑ Usability tests (formative-summative)
- ❑ Basic concepts
 - Research Question (RQ)
 - Hypothesis
 - Qualitative data
 - Quantitative data
- ❑ Data collection methods
 - Observation, interview, focus group
 - Lab and field experiments
- ❑ Measurement
- ❑ Data analysis
 - Exploratory
 - Confirmatory
- ❑ Research-informed System Design



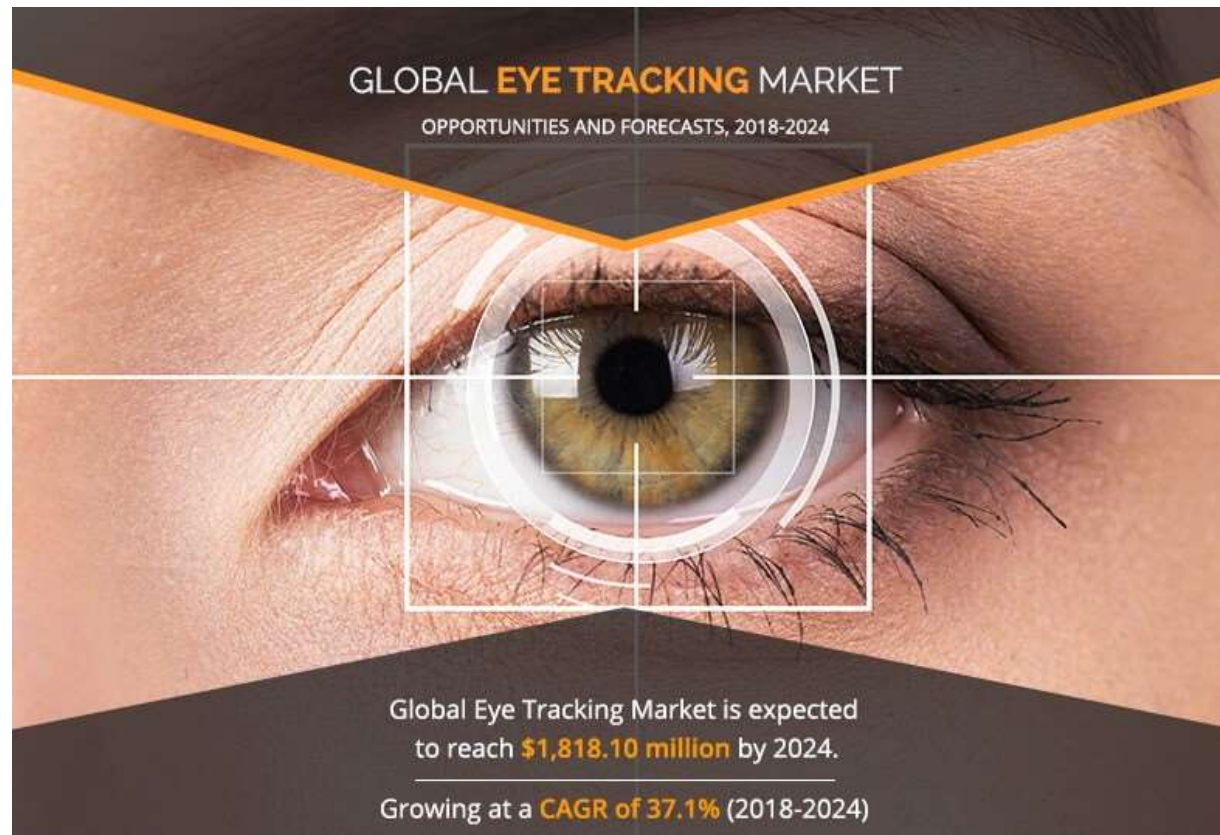
What you'll learn the coming two weeks

- ☐ Usability testing
- ☐ Experimental design in HCI
- ☐ Plan and set up an experiment
- ☐ Interpret results
- ☐ Statistics in HCI



Usability tests. Why?

After you graduated, you want to do a start-up in machine learning to analyze *visual pointing* areas. You analyze eye trackers and design algorithms to optimize accuracy. You go to market but don't sell. After a year, you are out of business. How come?



Usability tests. Why?

After you graduated, you want to do a start-up in machine learning to analyze *visual pointing* areas. You analyze eye trackers and design algorithms to optimize accuracy. You go to market but don't sell. After a year, you are out of business. How come?

$$x(t) := \exp \left\{ \int_0^t \left[\frac{\alpha^2(s)}{2} - a(s) \right] ds - \alpha(s) dB(s) \right\} \\ \times \left[1/N_0 + \int_0^t b(s) \exp \left\{ \int_0^s \left[a(\tau) - \frac{\alpha^2(\tau)}{2} \right] d\tau + \alpha(\tau) dB(\tau) \right\} ds \right]; \quad (2.2)$$

then $x(t)$ satisfies the equation

$$dx(t) = \left[(\alpha^2(t) - a(t)) dt - \alpha(t) dB(t) \right] x(t) + b(t) dt. \quad (2.3)$$

Let $N(t) := 1/x(t)$, then $N(t) > 0$ and $N(t)$ is continuous and global on $t \in [0, \infty)$. By Itô's formula

$$\begin{aligned} dN(t) &= -\frac{dx(t)}{x^2(t)} + \frac{(dx(t))^2}{x^3(t)} \\ &= -\left[(\alpha^2(t) - a(t)) dt - \alpha(t) dB(t) \right] N(t) - b(t) N^2(t) dt + N(t) \alpha^2(t) dt \\ &= N(t) \left[(a(t) - b(t) N(t)) dt + \alpha(t) dB(t) \right]. \end{aligned}$$

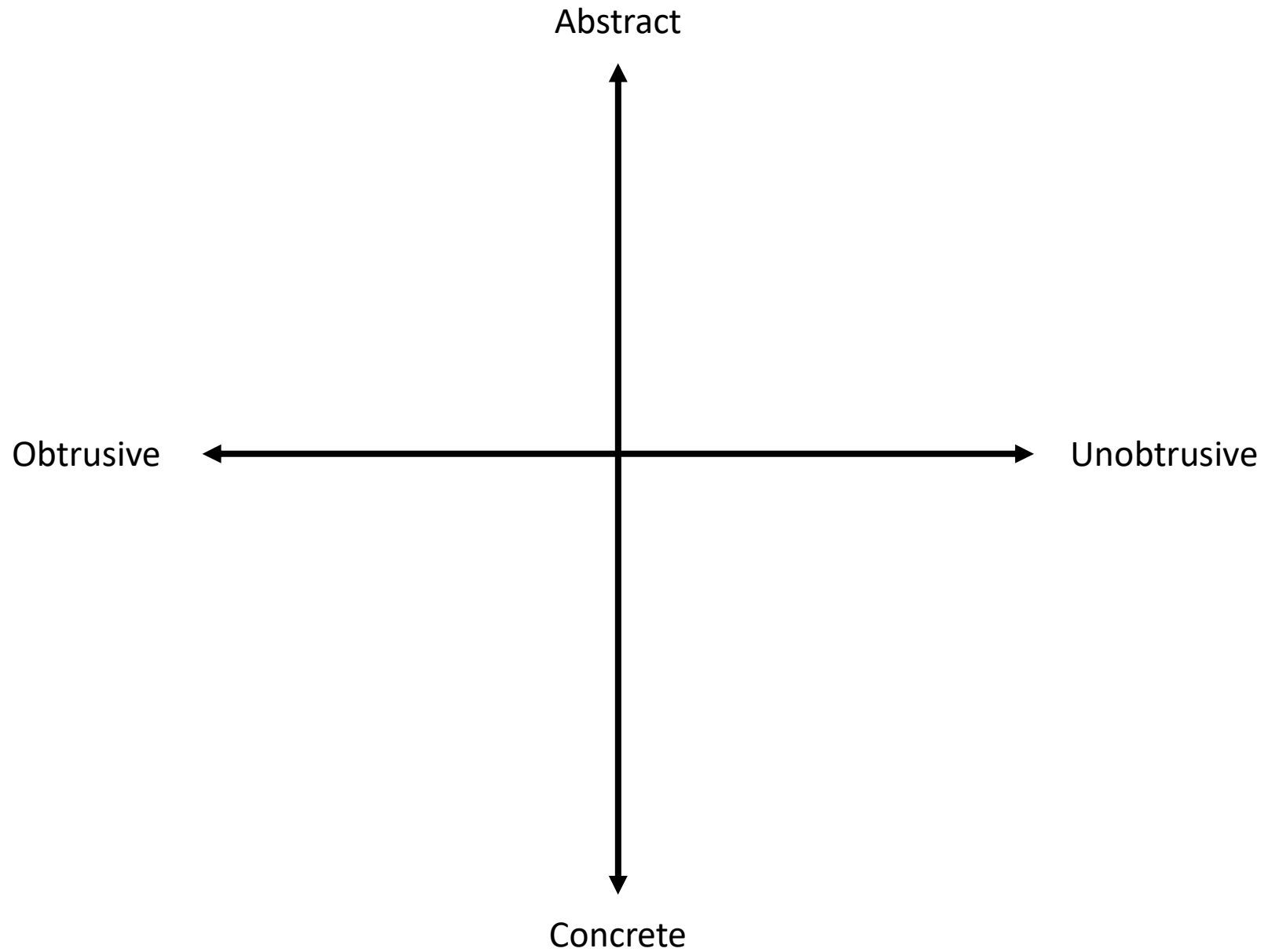
Your eye was blocked by algorithm

Usability tests. Why?

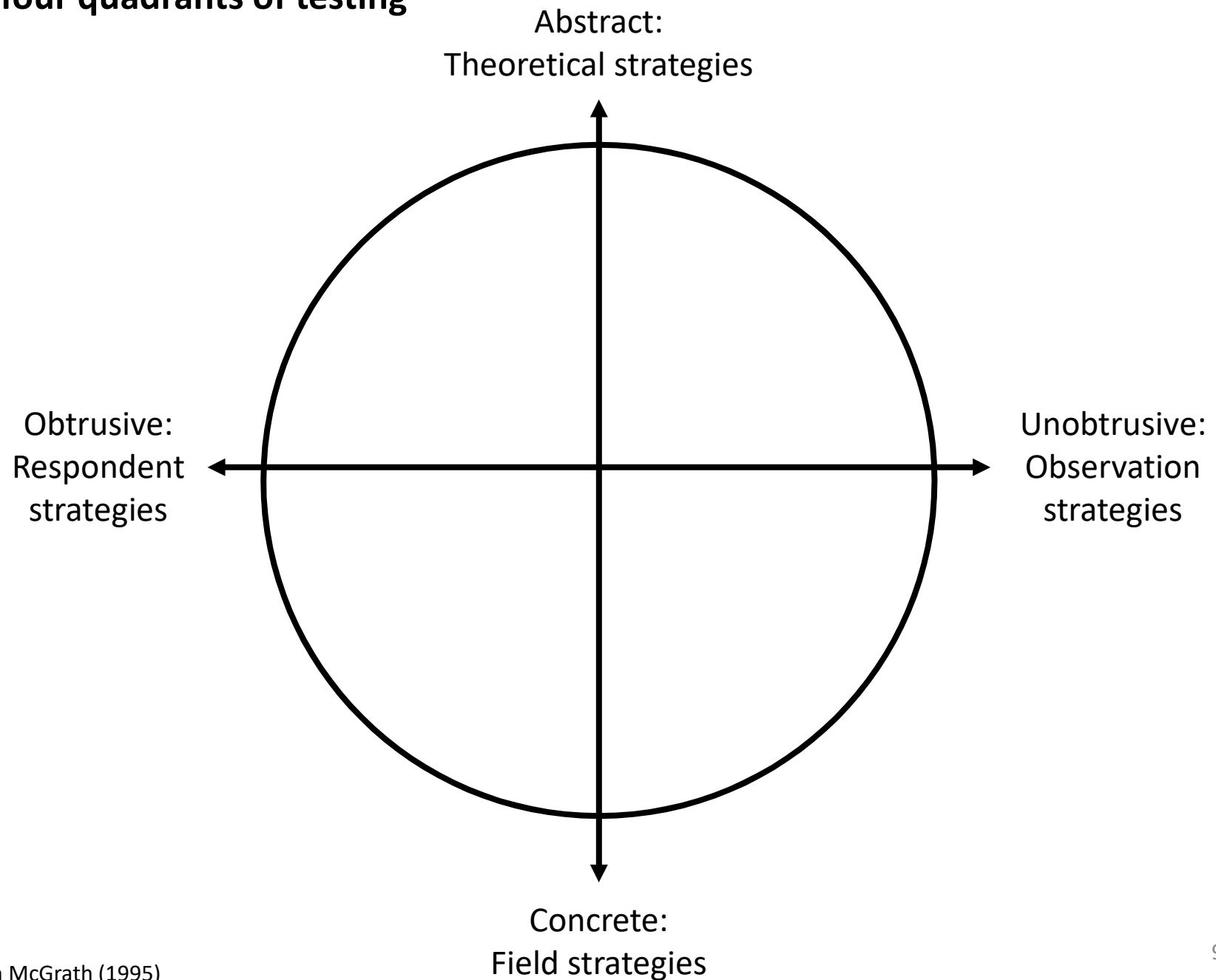
1. Whose problem are you solving? Is there a need for visual pointing? Or were you fascinated by machine learning and found a suitable application area for your algorithms?
2. You ignored Fitts' Law and did not do user-performance testing. To what degree does your system improve the current visual pointing systems? A little bit? Enormously? What does it do for time? Slower? Too fast?
3. You did not relate performance to usability. Perhaps you improved the accuracy but made the system less useful in other ways, for example, because the algorithm consumes much memory space or cannot be used offline
4. You did not measure user satisfaction as related to performance. Perhaps your client is OK with a more coarse resolution and a few misses every now and then. The need to change may be minimal
5. Without HCI, you may invest a lot of development time in improving some system, which makes your final product more expensive than the existing solutions with the risk that no one is waiting for it

Fitts' law states that the amount of time required for a person to move a pointer (e.g., mouse cursor) to a target area is a function of the distance to the target divided by the size of the target.

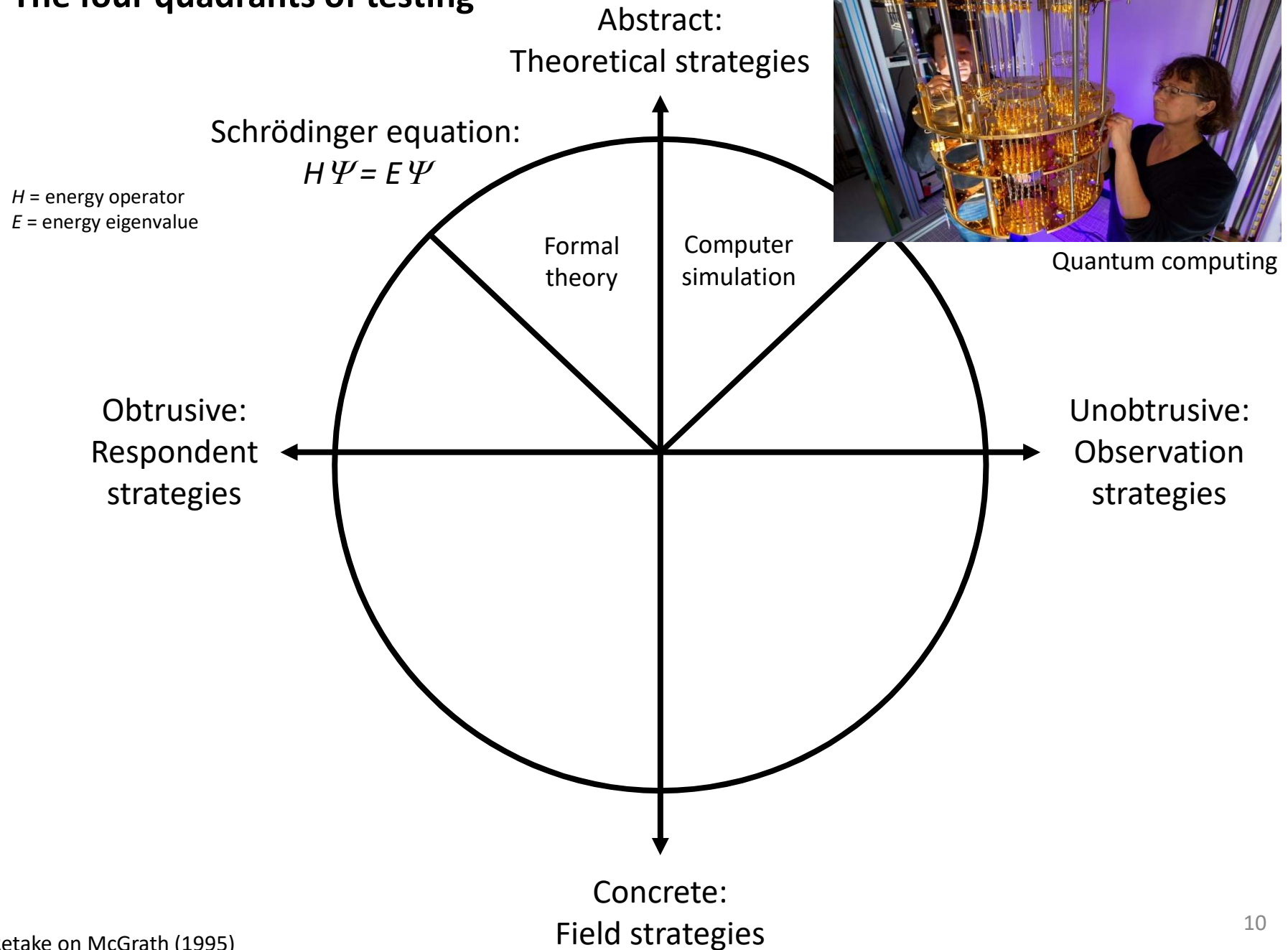
The four quadrants of testing



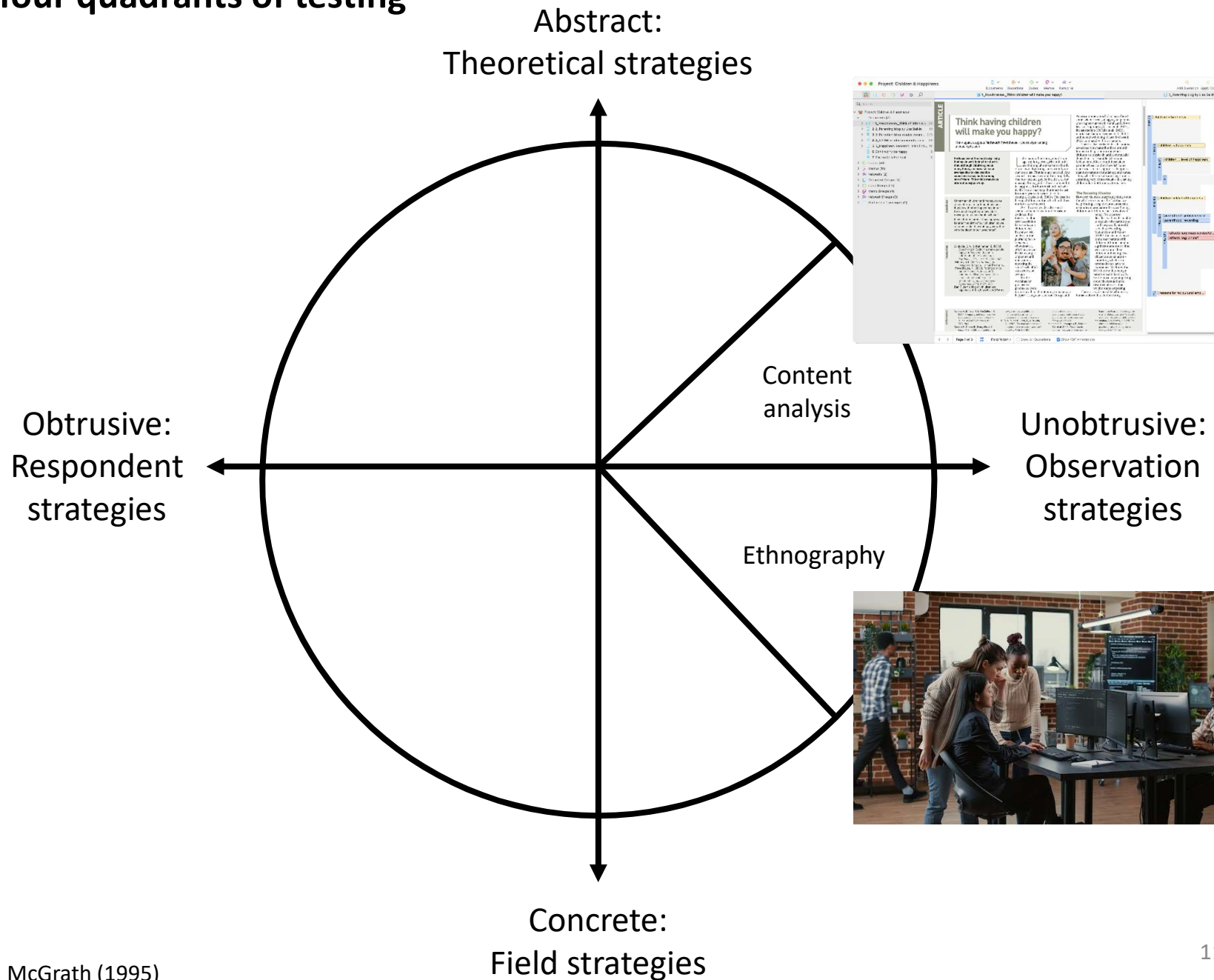
The four quadrants of testing



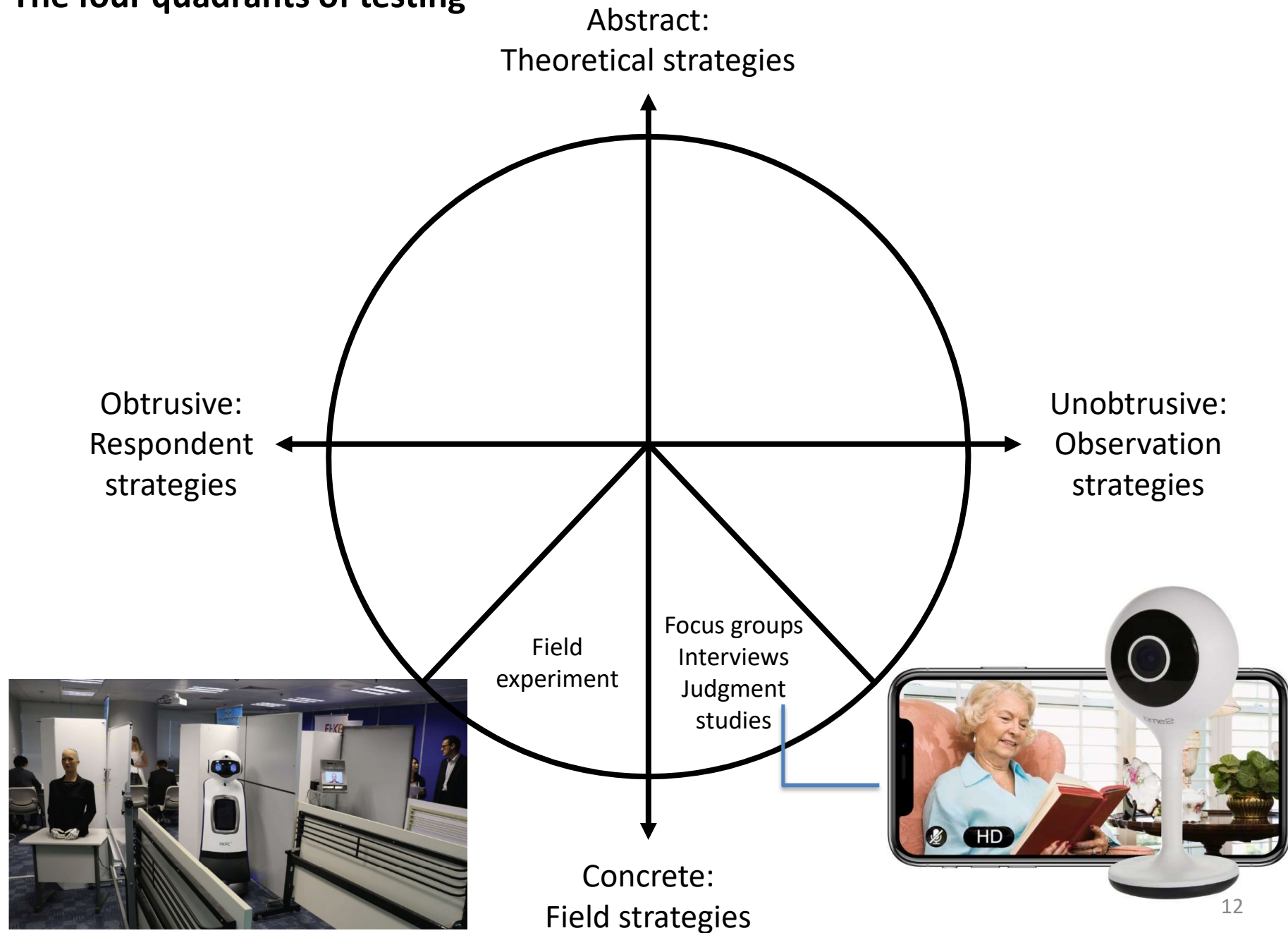
The four quadrants of testing



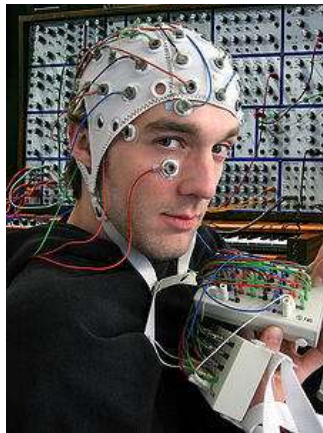
The four quadrants of testing



The four quadrants of testing



The four quadrants of testing



Obtrusive:
Respondent
strategies

Laboratory
experiment

Sample surveys

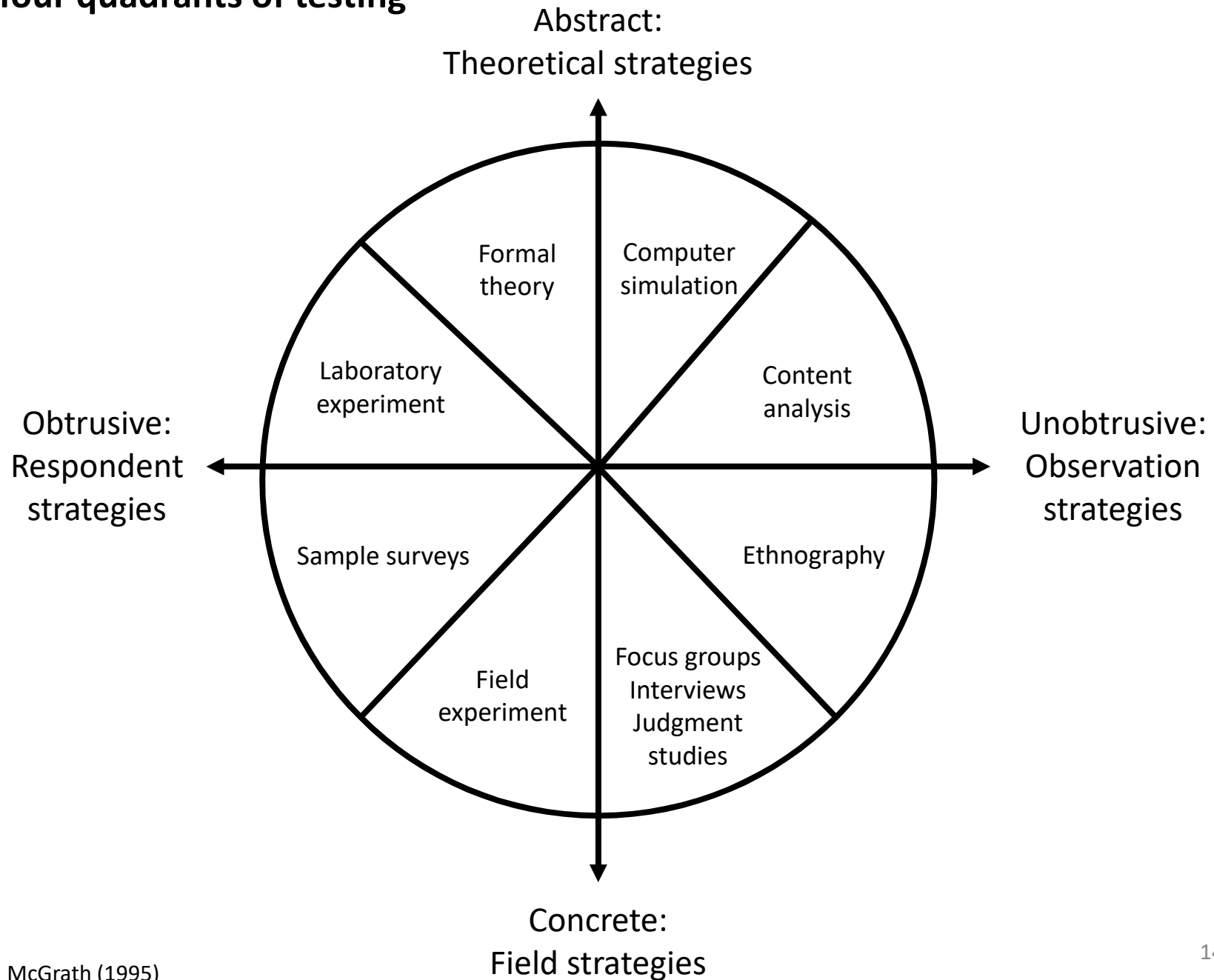
Unobtrusive:
Observation
strategies

Abstract:
Theoretical strategies

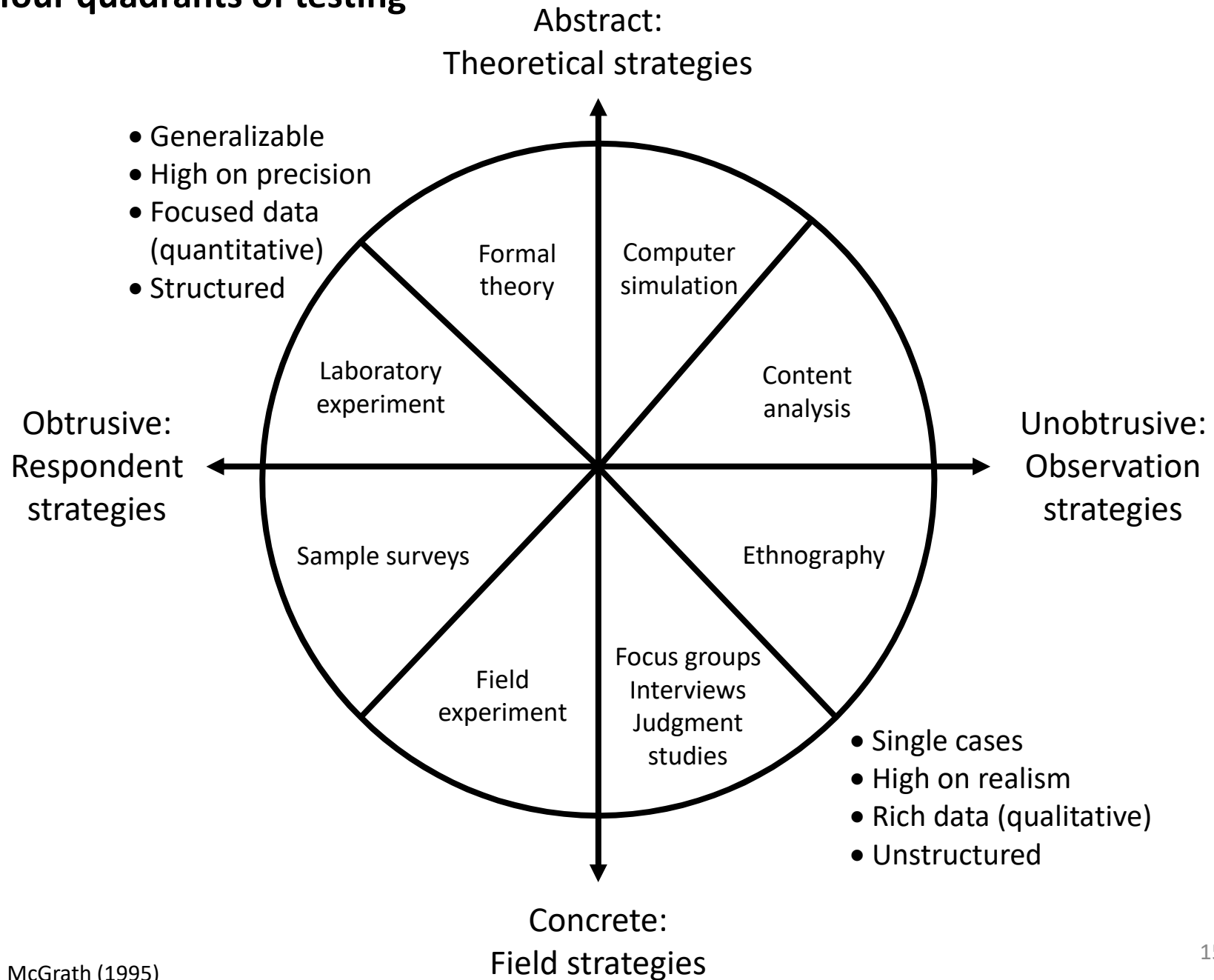
Concrete:
Field strategies



The four quadrants of testing



The four quadrants of testing



Research Question

No matter what, all research starts with a Research Question (RQ), which guides you through the whole research process. It is the thing you fall back on when you are stuck or are digressing; it keeps you on topic

❑ Formulating an RQ

- State the general topic
- Narrow down the topic through desktop research
- Question the existing answers
- Find the research gap
- Reformulate the RQ

What does it mean to...

to understand humans

human language

not just syntax and semantics

multimodal interaction

pragmatics of language use



Science = know why:

knowledge for its own sake
(don't ask about usefulness)

Engineering = know how:

knowledge to make things
work (e.g., a program to
understand humans)

Hypotheses

- ❑ Derived from the RQ, a hypothesis is a **precise problem statement** about a topic (e.g., about a physical system, a computer system, human experience)
 - States the expected statistical relationship between groups of data
 - The statement has to be precise and state what we will be measuring!

Not precise enough (what does 'better' mean here?):

- Humans are better than AI
- iPhone is better than Blackberry

A little better:

- Humans are more intelligent than AI
- The iPhone keyboard is faster to type on than the Blackberry keyboard

Blackberry had a physical keyboard



Most precise:

- Humans have higher scores on standardized IQ tests than does AI
- Users take less time (measured in seconds) to enter in text messages on the iPhone keyboard than on a Blackberry keyboard

The Null and the Alternative hypothesis: H_0 and H_A

□ The hypotheses we formulate are commonly about expected effects

- Statements about effects are statements about differences, about change:
 - Before-after treatment (medication, washing powder, robot intervention)
 - With FinTech, handling your finance is easier than ever
 - My GPU is faster than yours
 - Assassin's Creed provokes more intensive emotions than Mario Kart
- That means that there must be at least two sets of data (else, nothing to compare)
- It also means that there must be at least two hypotheses:

H_0 There is no effect

The Null hypothesis (no detectable differences)

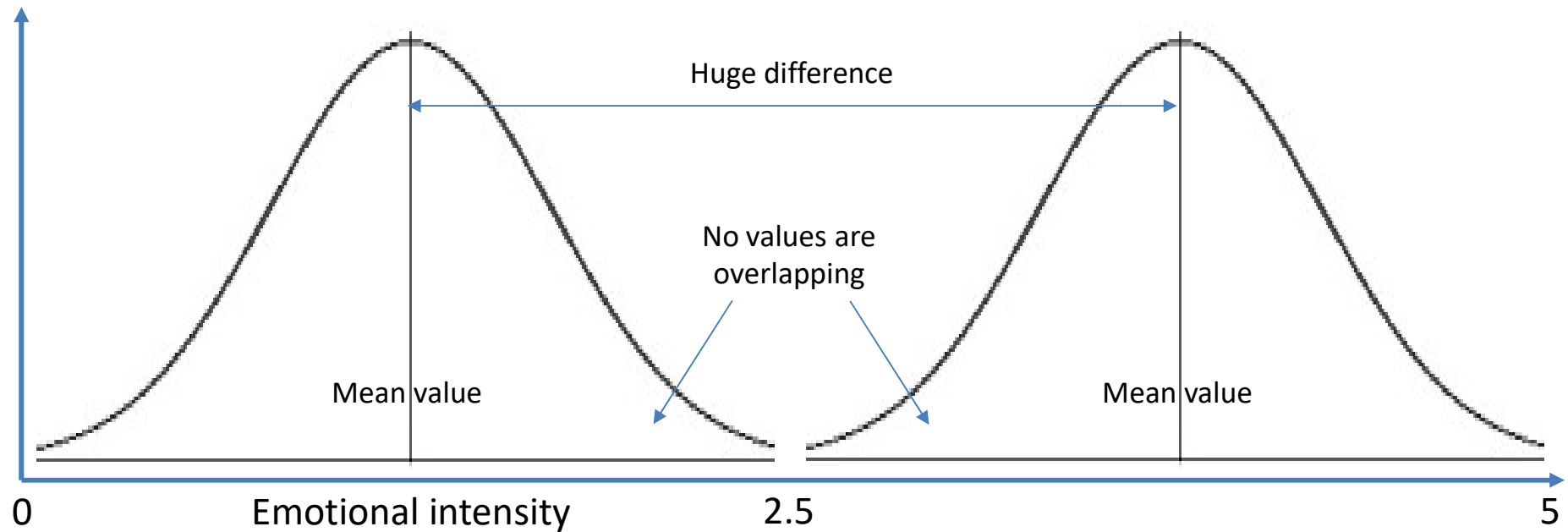
H_A There is an effect

The Alternative hypothesis (also called H_1)

- H_0 predicts that differences between two sets of data are statistically not significant
- H_A predicts that differences between two sets of data are statistically significant

Later we will discuss what 'significant' means as it is the researcher deciding when an effect counts or not, based on current knowledge, test theory, mathematical approach, application value, and much more

Number of people (probability that a value occurs based on frequency)

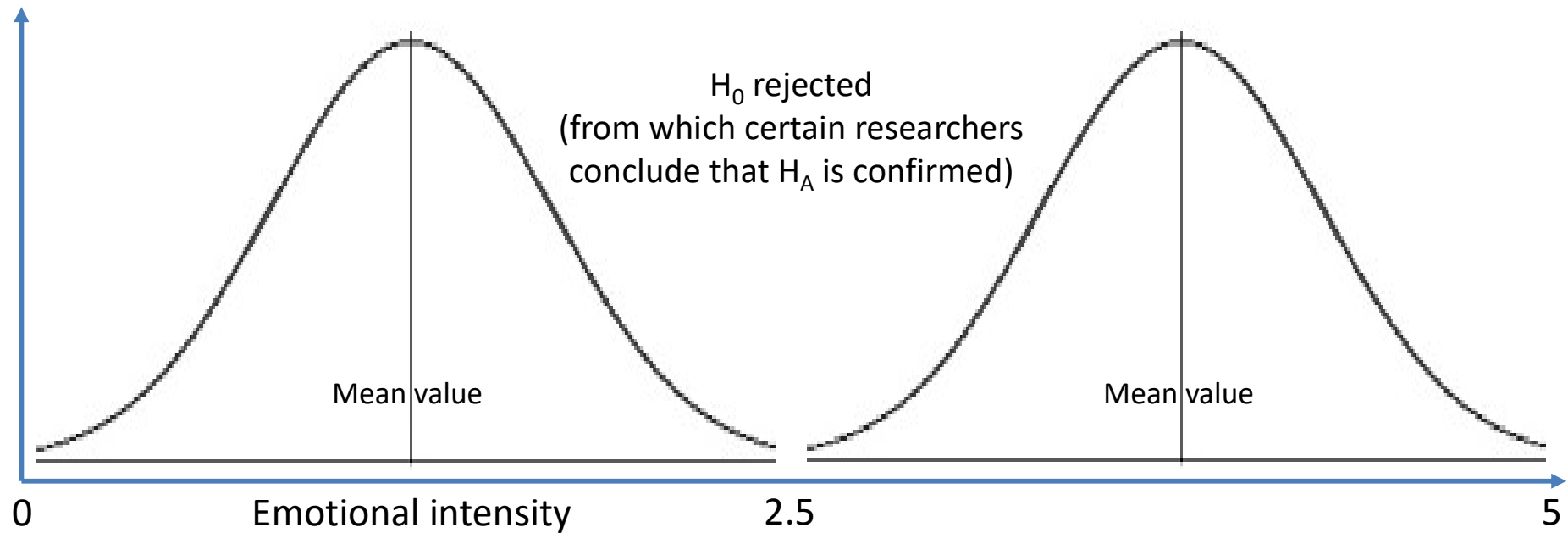


Mario. Mario Kart 8, Nintendo (2014)



Basim. Assassin's Creed Mirage, Ubisoft (2023)

Number of people (probability that a value occurs based on frequency)

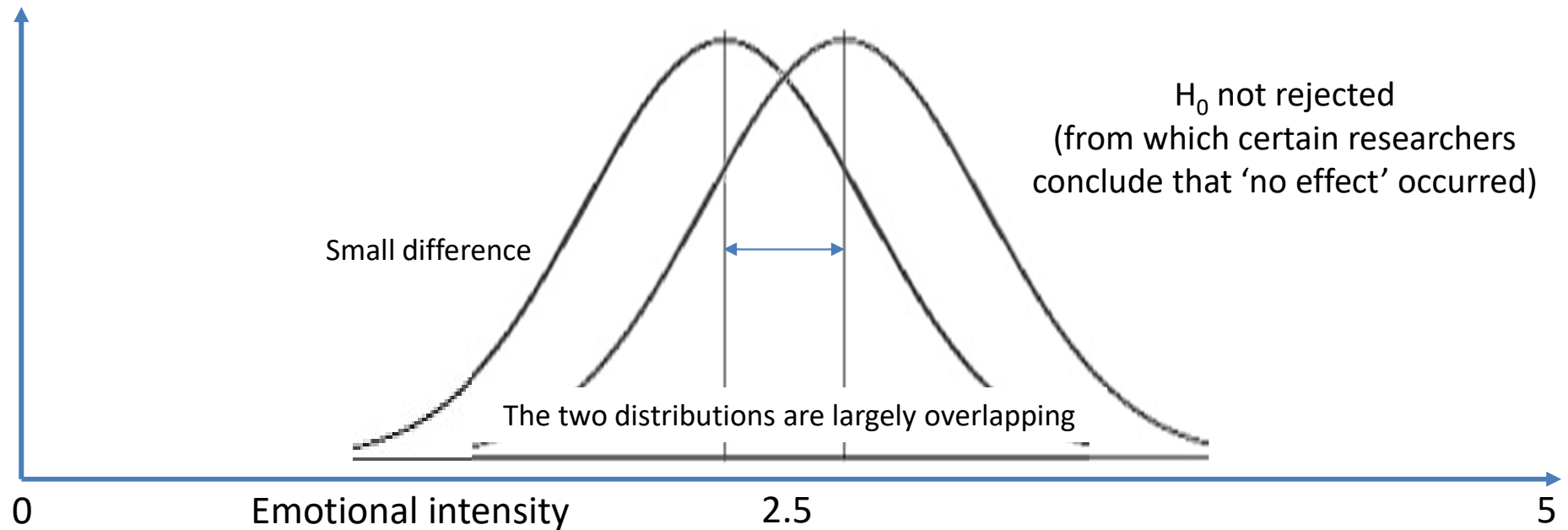


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Number of people (probability that a value occurs based on frequency)



Mario. Mario Kart 8, Nintendo (2014)



Basim. Assassin's Creed Mirage, Ubisoft (2023)

The Null and the Alternative hypothesis: H_0 and H_A

- ❑ The Null hypothesis states the opposite of the expected statistical relationship that the Alternative predicts
- The Null is conservative, favors current knowledge unless overwhelming evidence systematically refutes current beliefs

H_0 Users experience the same or even less intense emotions (on a scale from 0 to 5), playing Assassin's Creed compared to playing Mario Kart

H_A Users experience more intense emotions (on a scale from 0 to 5), playing Assassin's Creed than playing Mario Kart

H_0 Users take the same or more time (measured in seconds) to enter in text messages on the iPhone keyboard compared to a Blackberry keyboard

H_A Users take less time (measured in seconds) to enter in text messages on the iPhone keyboard than on a Blackberry keyboard

If the observed data show that it is significantly unlikely that there is no effect (so there is difference into the expected direction), then the Null hypothesis is rejected, and the Alternative hypothesis is accepted

Similar to a criminal trial: The defendant is assumed to be innocent (the Null hypothesis) until proven guilty (Null is rejected) beyond a reasonable doubt (to a statistically significant degree). This is a fallacy (i.e. an *appeal to ignorance*) but that's how frequentist science works

Independent variables or ‘fixed factors’

- ❑ The condition under which a task is performed is the *independent* variable (e.g., game type, telephone brand, learning with or without VR support)
 - Factors that are systematically varied, manipulated, and controlled by the researcher
 - Determines a modification to the conditions that people or physical objects undergo
- ❑ May also arise from demographic data (e.g., education level, gender, or age)
 - “Women remember more items after 20 minutes browsing the web than men”
 - “Older adults report higher satisfaction with a companion robot than do youngsters”



Mario. Mario Kart 8, Nintendo (2014)



Basim. Assassin's Creed Mirage, Ubisoft (2023)

Dependent variables or 'measures'



☐ Measurements (often quantitative)

- The things that we set out to measure/observe (e.g., temperature, time, a rating)
- Demonstrate the effect of the independent variables (e.g., fun while gaming)
- Depends upon the independents as well as the user's behavior or performance

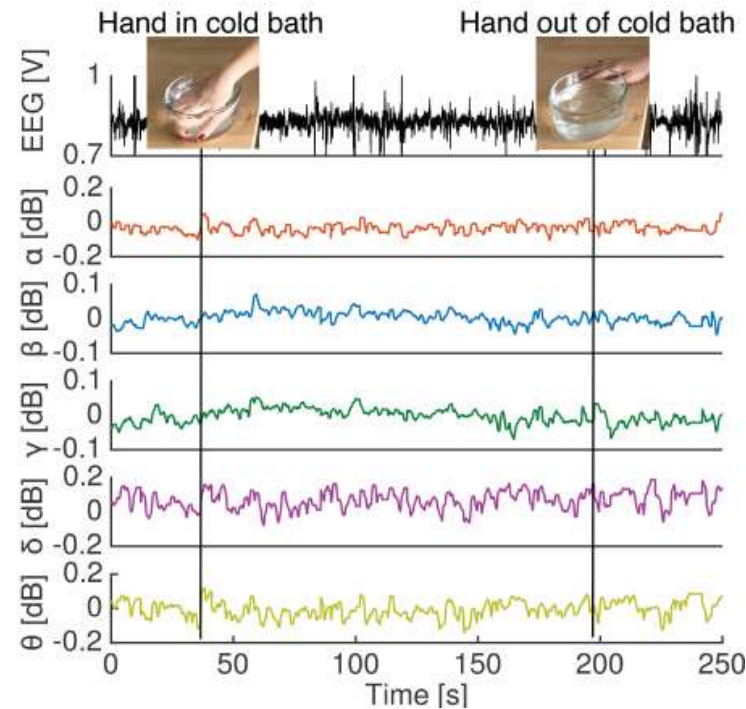
☐ Properties

- Readily observable and readily related to some scale of measurement
- Stable and reliable so they do not vary under constant (experimental) conditions
- Sensitive to the effects of the independent variables

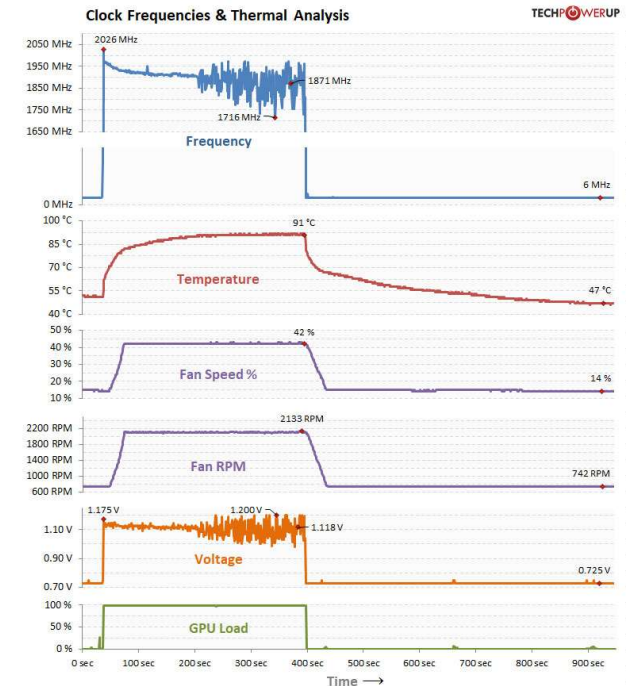
Examples: Number of errors made, time taken to complete a given task, time taken to recover from an error, attitude indicated on a rating scale

A dependent variable is the variable that changes as a result of the independent variable manipulation. It's the outcome you're interested in measuring, and it "depends" on your independent variable

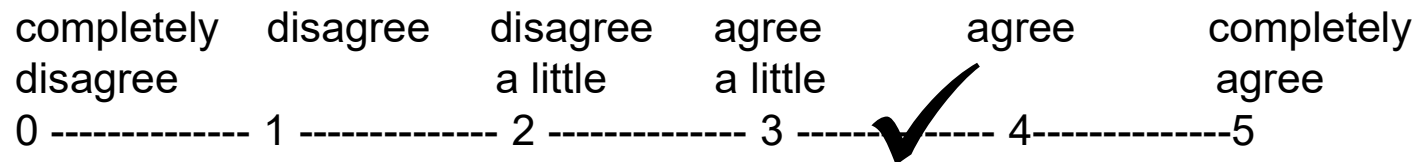
Dependent variables or 'measures' (some examples)



AMD Radeon RX 5700 XT



The interaction with Basim Ibn Ishaq from Assassin's Creed was exciting



Dependent variables: qualitative

❑ Qualitative data

- Written web reviews
- Camera footage
- Sound bites
- Observation protocols



“ This is a wonderful easy to use, VERY EASY to use service, both for your client as well as staff. I h...

Angled North

[view post](#)



“ We have been using Repuso for the past 8 months and it was absolutely brilliant, the team is s...

Seb Marouani

[view post](#)



Social reviews
Boost Conversion
Use Repuso! Create your first widget now!

“ Use in your site what other say about your company en #socialmedia... check @getrepu...

Germán Castaño

[view post](#)



“ Exactly what I was looking for our e-commerce sites. Easy to set up and simple to use. Extra credits t...

Ivar Arulaid

[view post](#)



“ What's more important than showing current customers and (more importantly) potential cust...

Greg Watson

[view post](#)



[More posts...](#)

Powered by *Repuso*

Dependent variables: quantitative

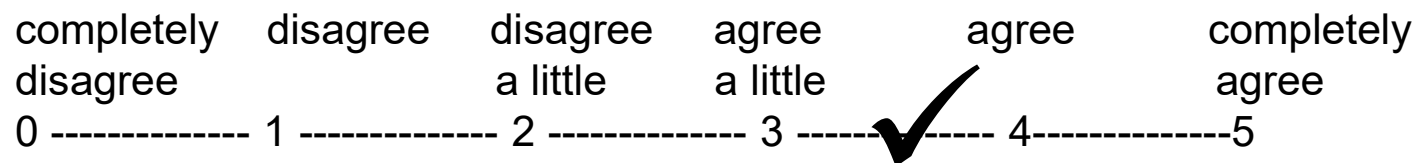
❑ Quantitative data

- Time: Easy to measure, suitable for statistical analysis (e.g. learning time, task completion time)
- Error rate (frequency of errors in a certain part of a system): Shows where problems exist within a system. May indicate cause of difficulties
- Patterns of system use (frequency of use of a certain part of the system):
Study the patterns of use. Preference and avoidance of particular sections
- Patterns of request for help (frequency of access to on-line help system):
Study what users find confusing

Confusion is the top 1 undesired effect of any application that you make!

- But also rating scales:

The interaction with Basim Ibn Ishaq from Assassin's Creed was exciting

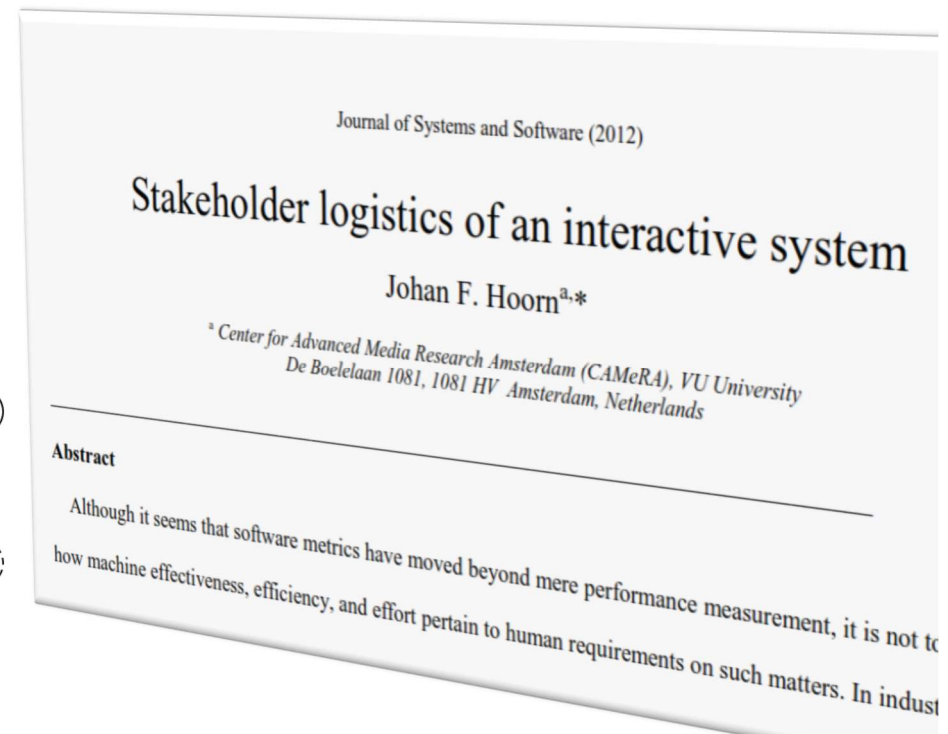
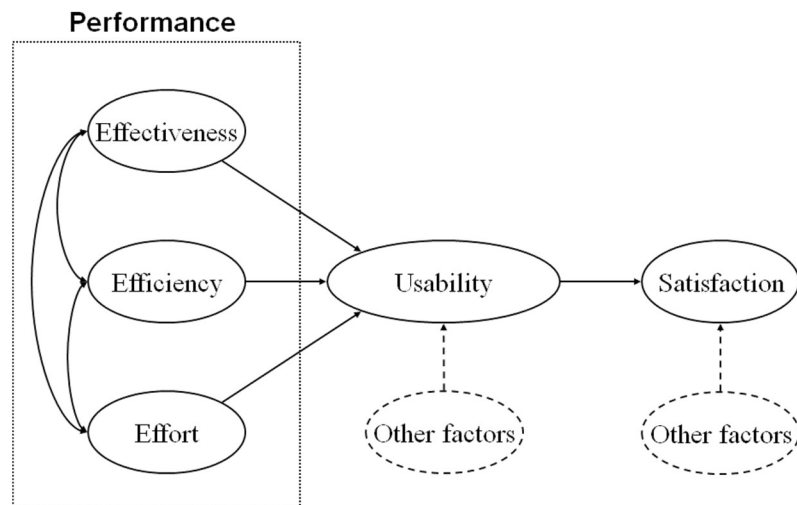


Quantifiable performance data in HCI

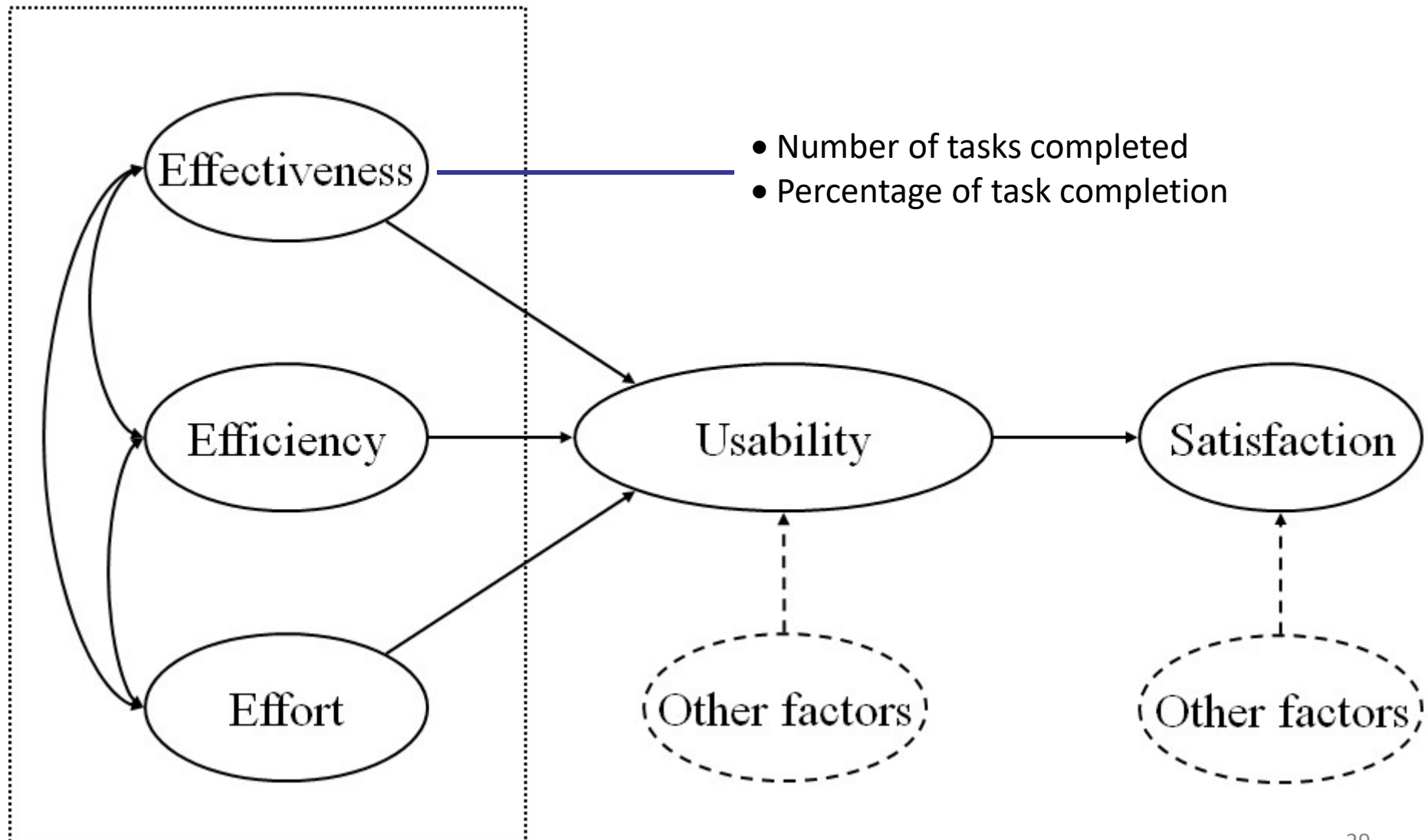
❑ Situations in which numbers are useful

- Effectiveness: Successful task completion rate
- Efficiency: The trade-off between speed and accuracy
 - Time requirements for task completion
 - Compare two systems on speed or number of errors
- Effort: Difficulty, ease of use but also energy consumption

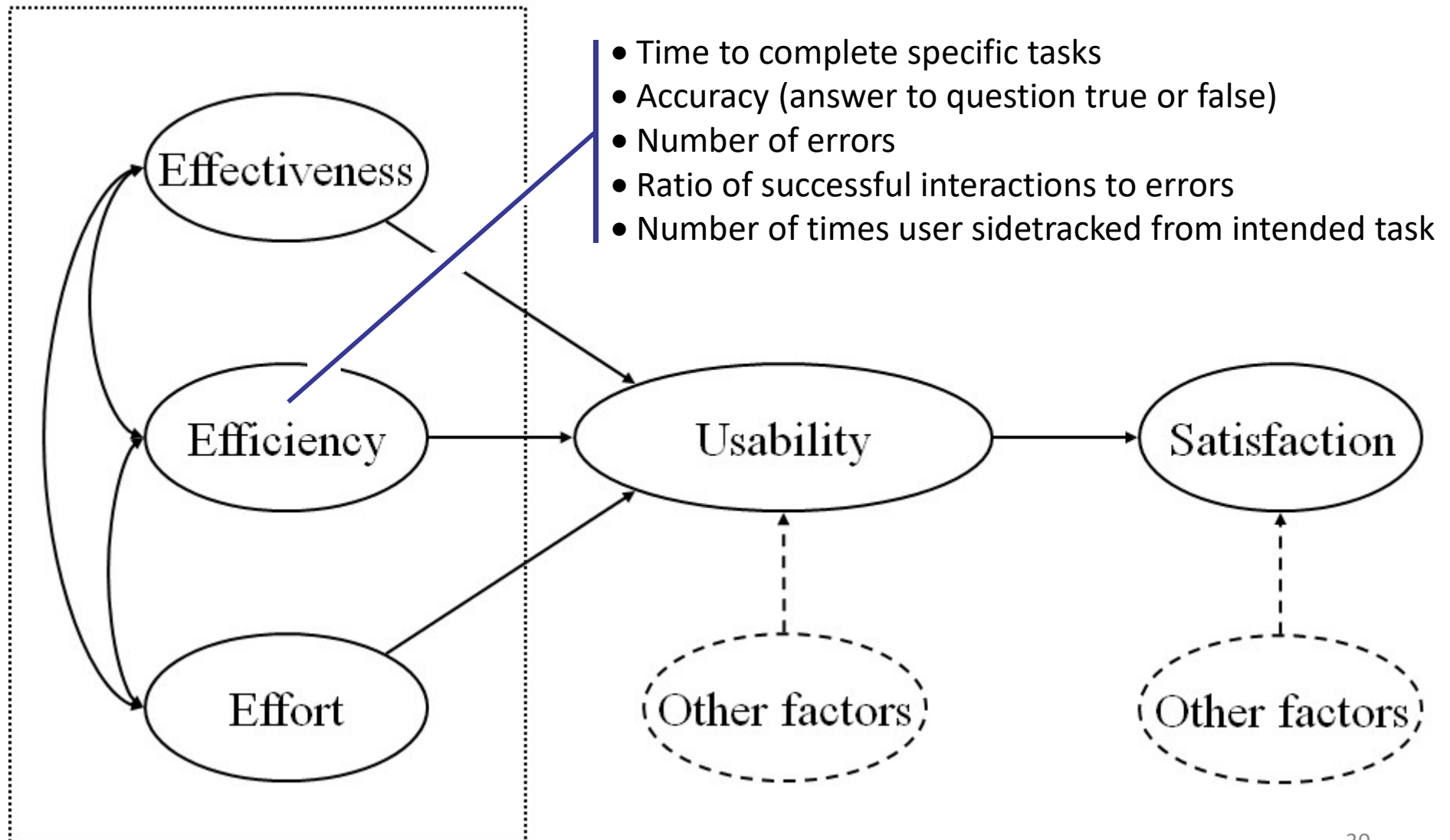
These are typical performance measures



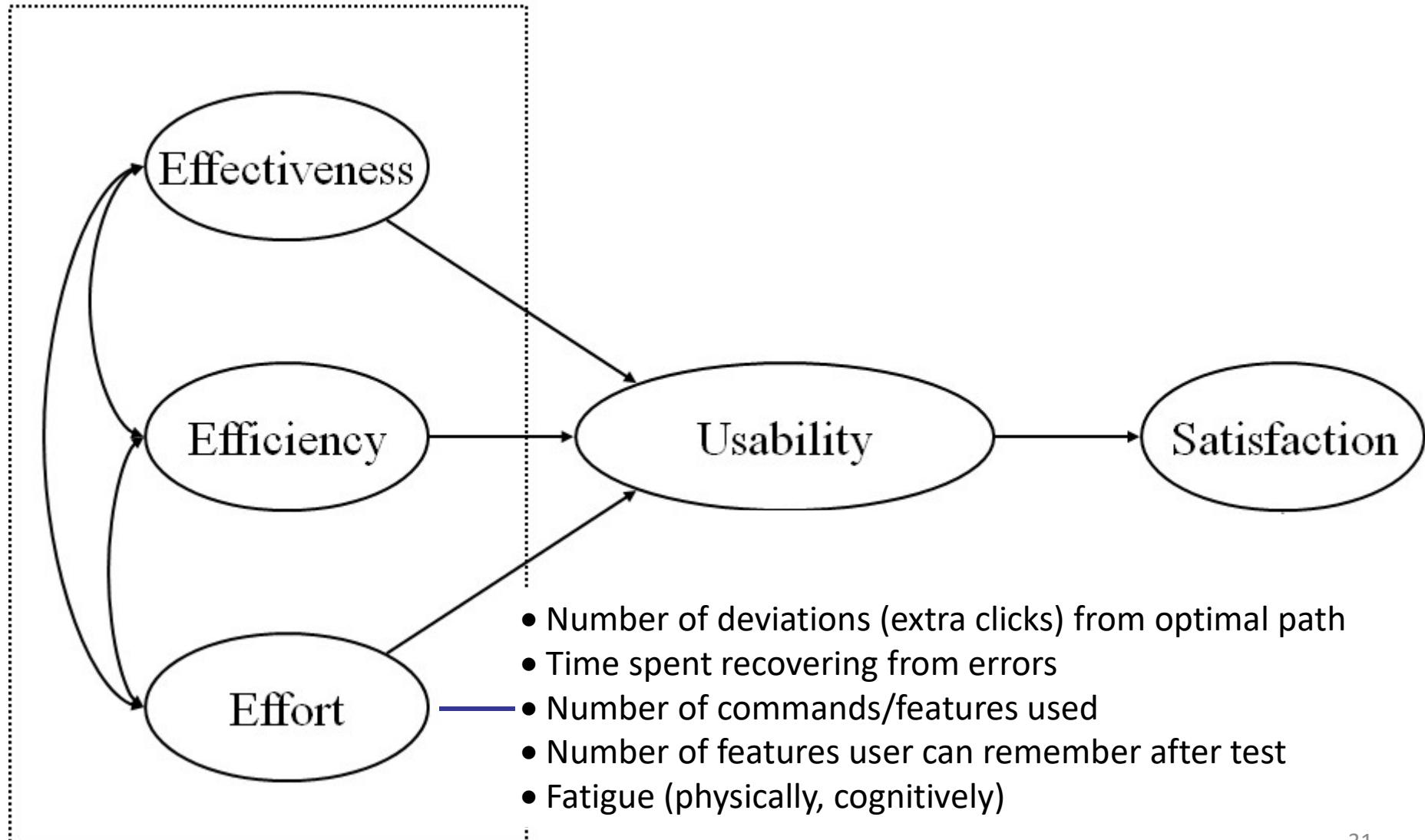
Commonly-used measures in HCI



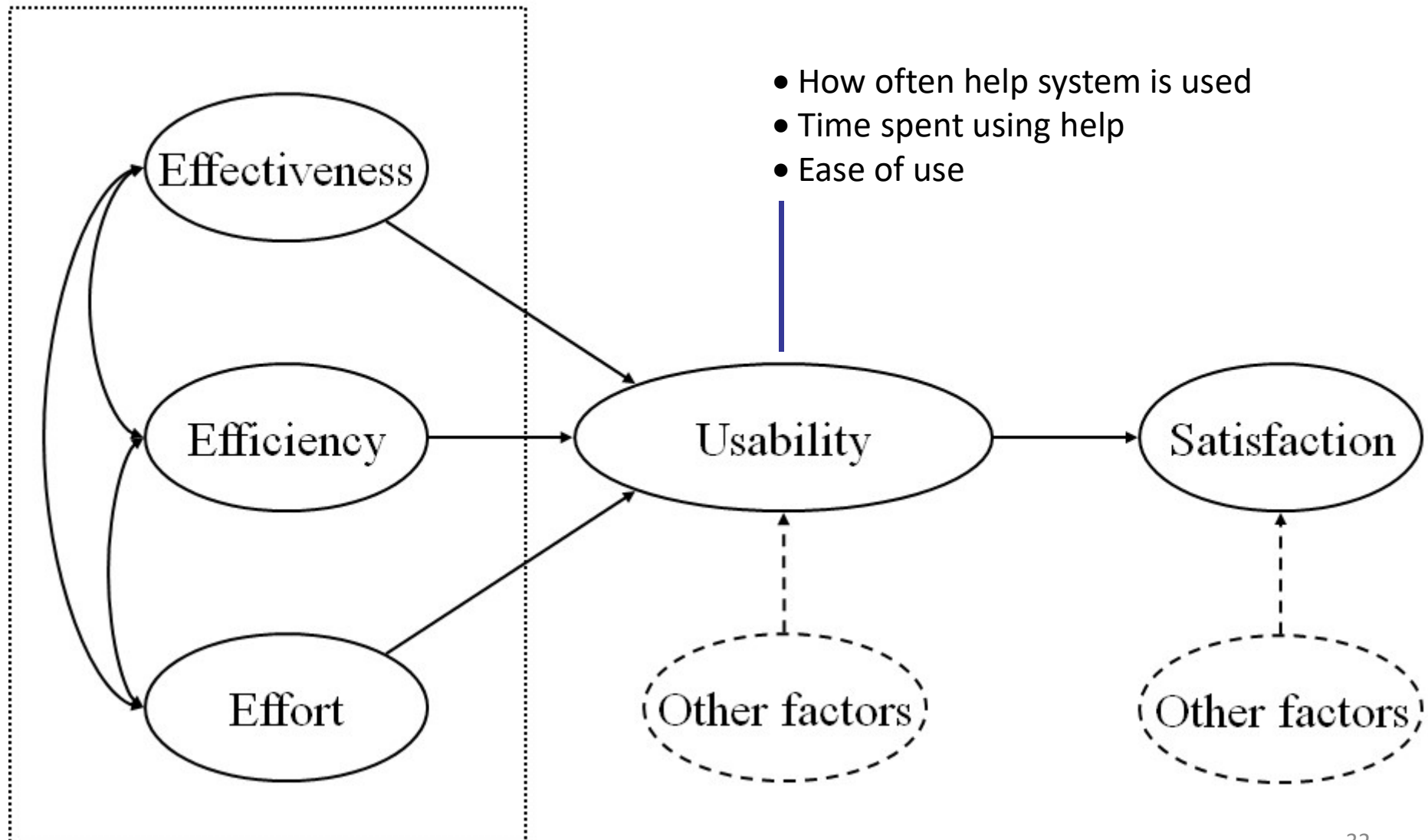
Commonly-used measures in HCI



Commonly-used measures in HCI



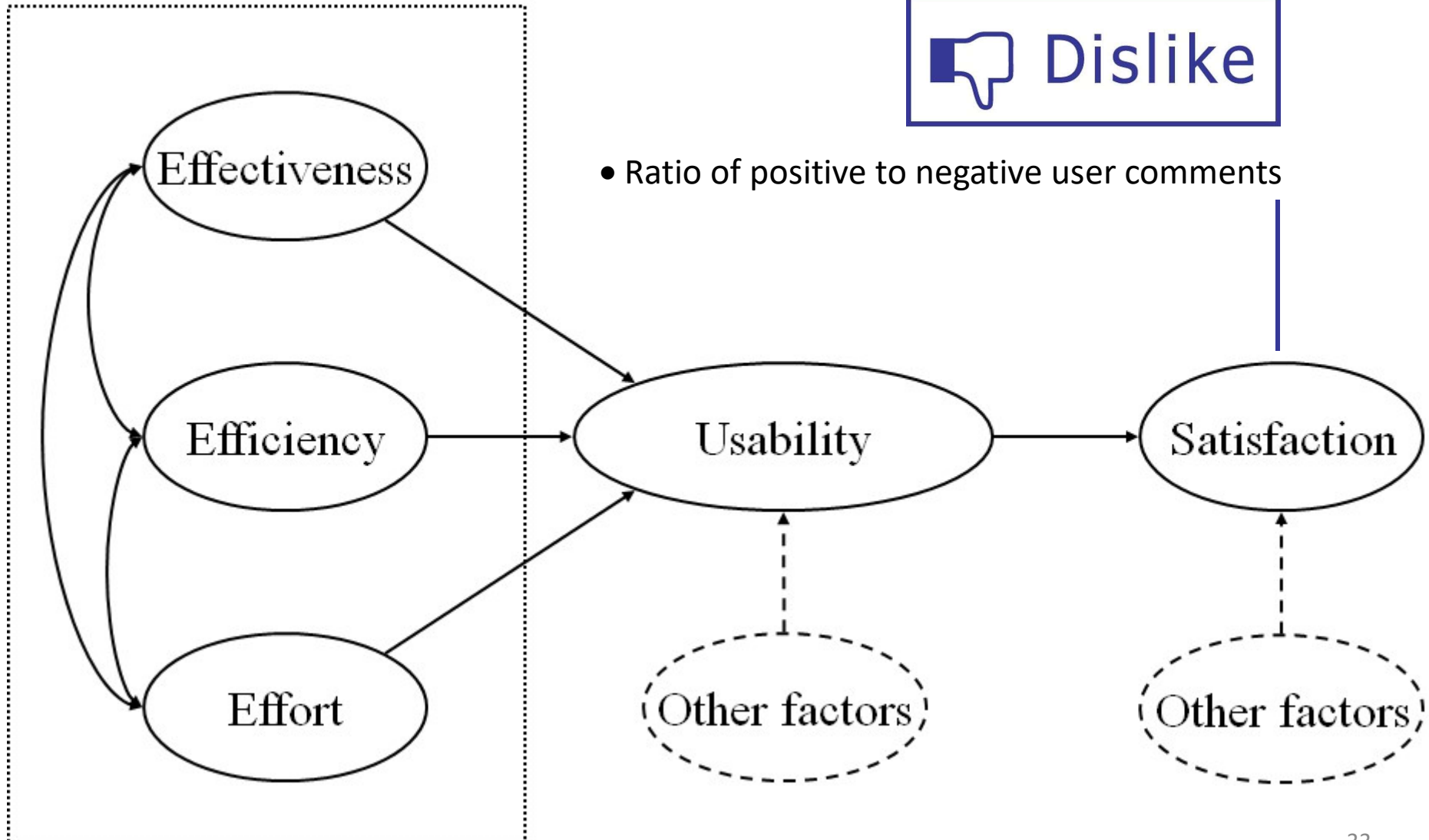
Commonly-used measures in HCI



Commonly-used measures in HCI



- Ratio of positive to negative user comments



Dependent variables: qualitative, quantitative

Tests may generate qualitative (i.e. strings) and quantitative data (i.e. integers)

☐ Qualitative evaluation

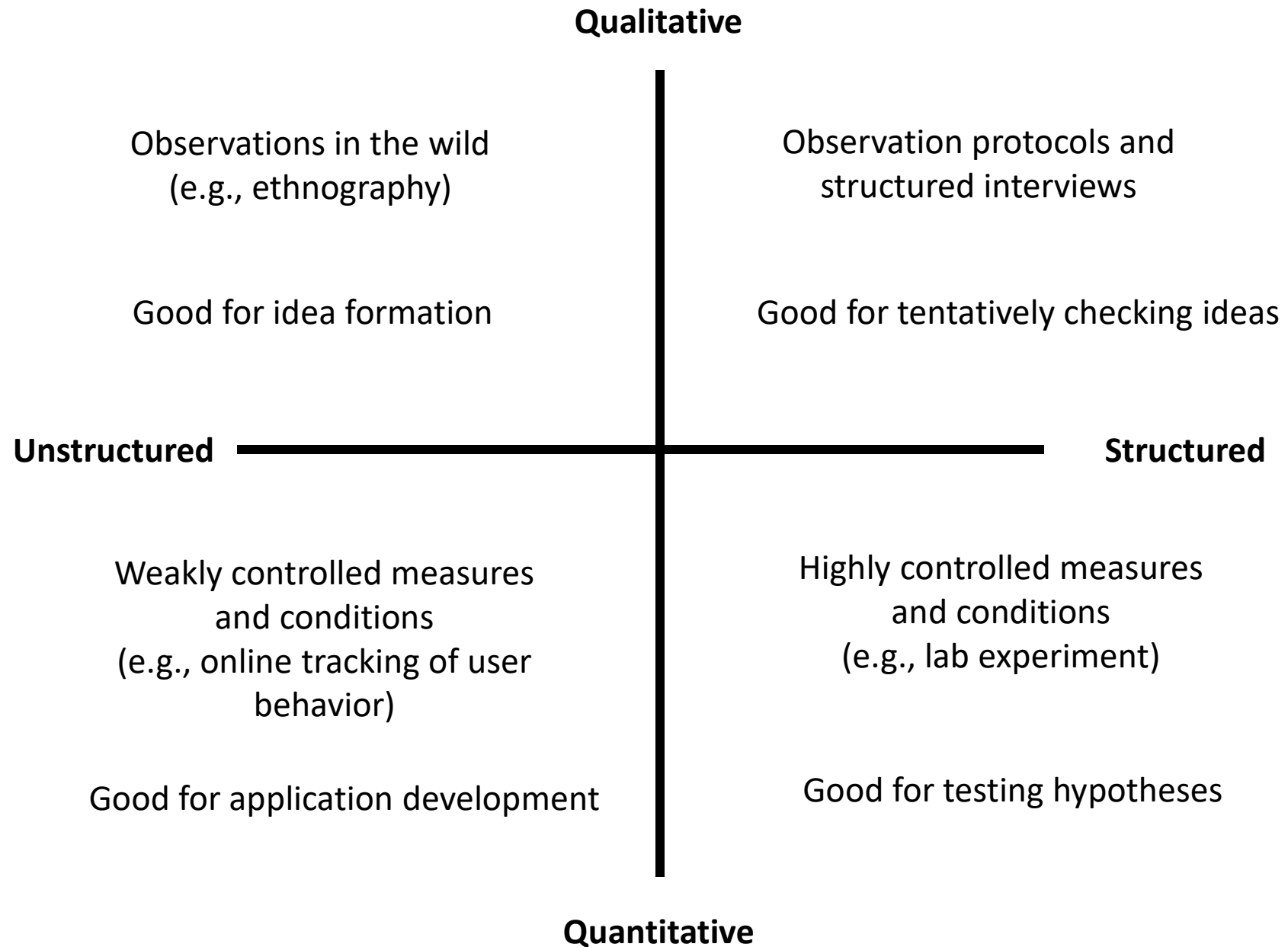
- Measurements are more *subjective* and *relative* (anecdotal)
- Gives us cases, data are *less structured*, more diverse, or ‘rich’

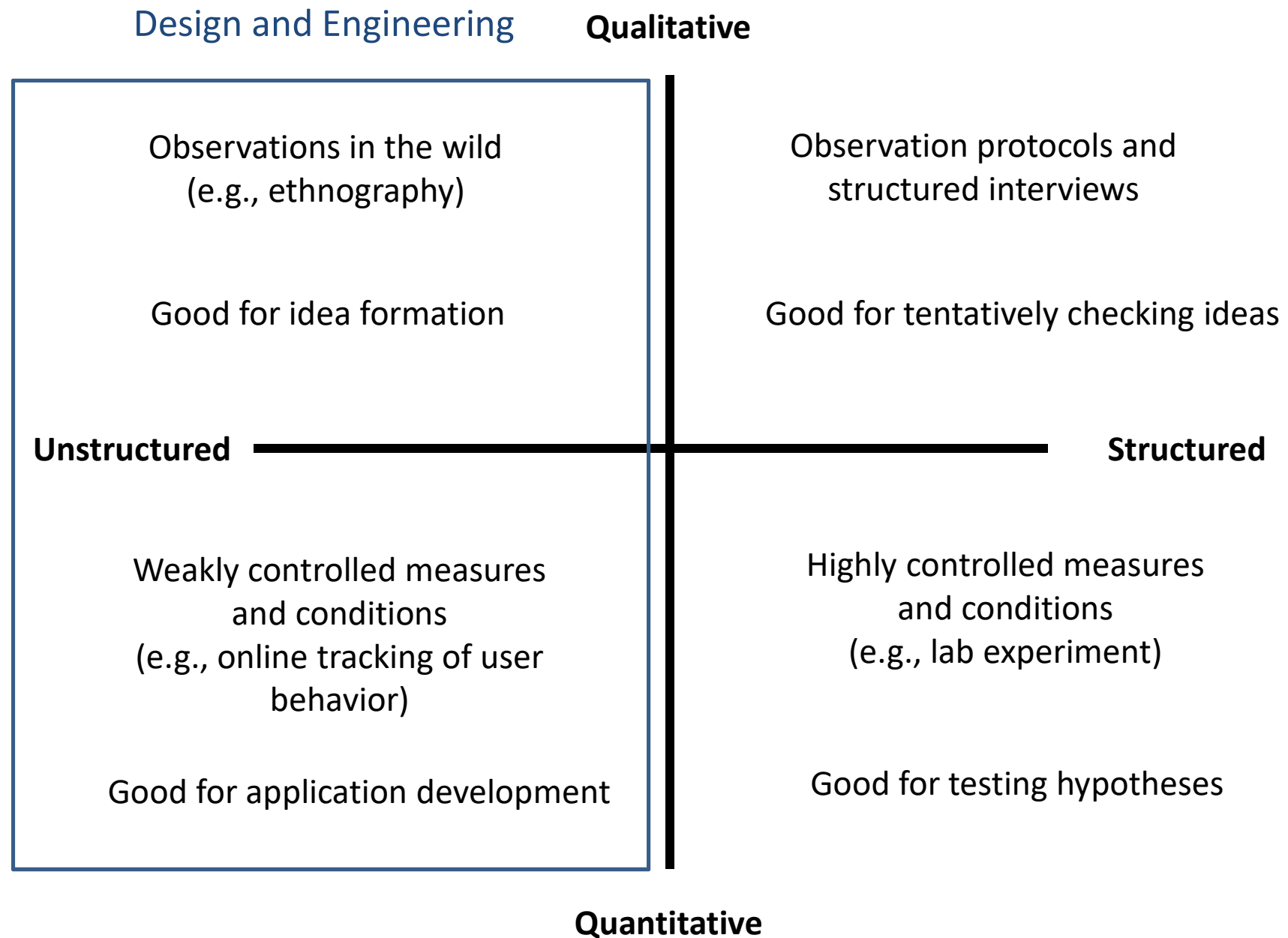
☐ Quantitative evaluation

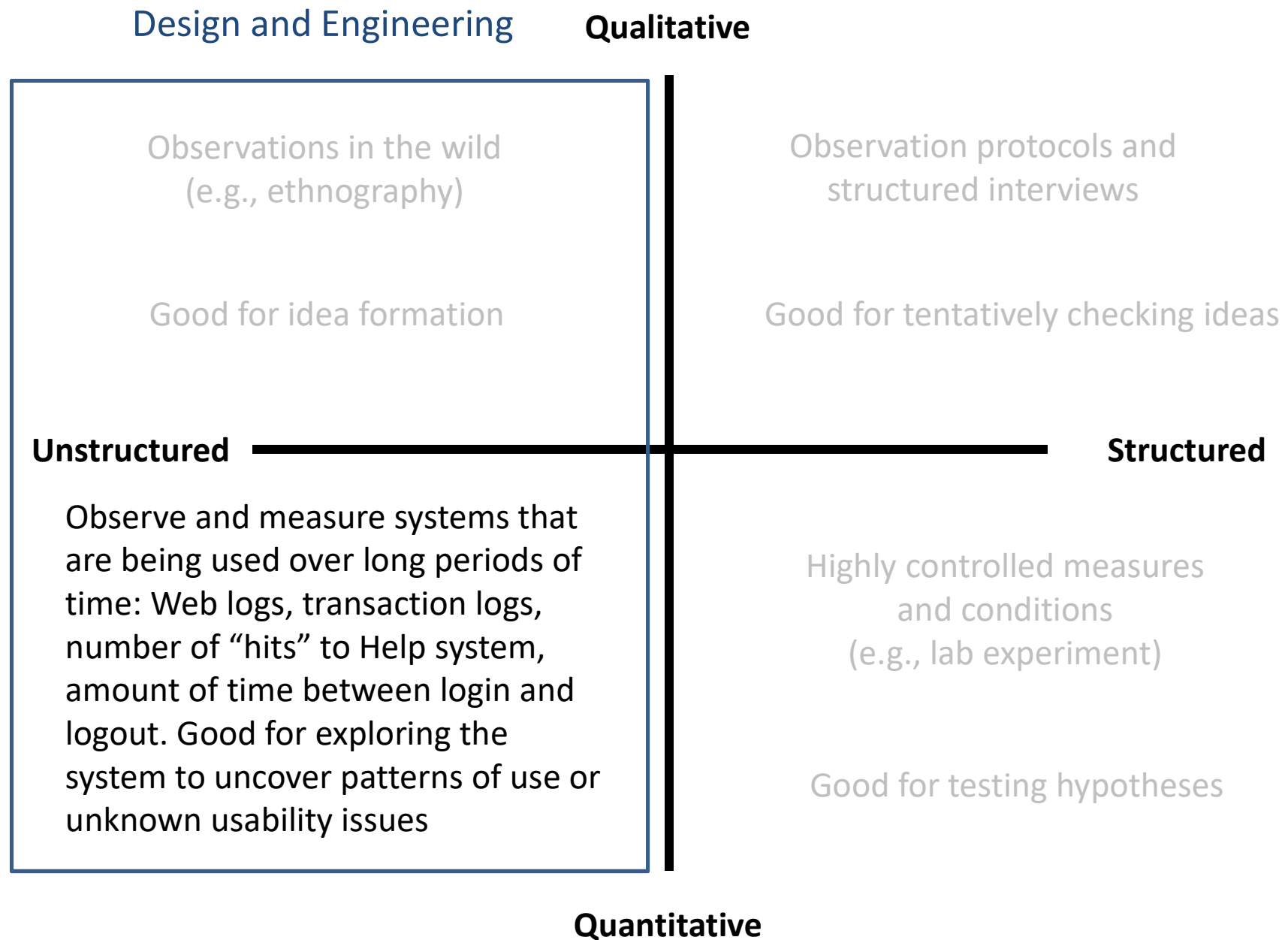
- Measurements are more *precise* and *less subjective*
- Gives us better *controllable* results, usually more *structured*, more *uniform*, but less ‘rich’
- More formal data, allowing for statistical analyses

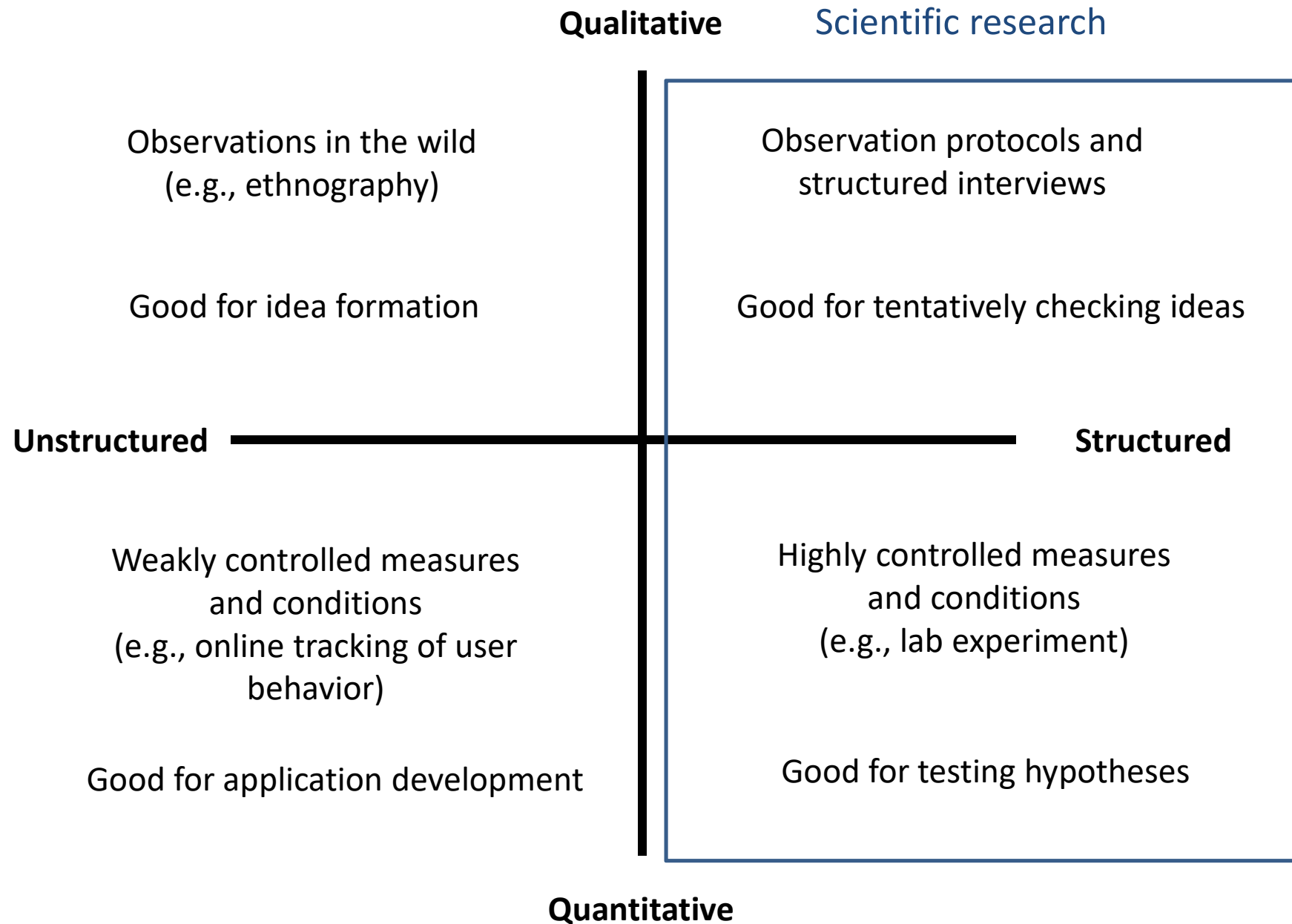
Typical conclusions after statistical analysis:

“On average, it took users 30% less time to finish a transaction, using FinTech.”





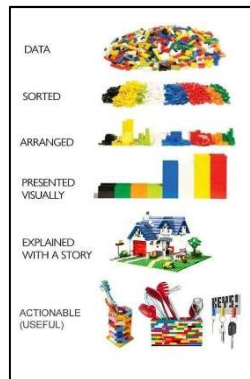




Design and Engineering

Keep in mind that adding more structure to data is adding more conceptual knowledge, which is 'theory'

Whether intuition, tentative knowledge, layman's theory, common sense, or best practice; all of that is based on implicit and thus theoretical assumptions



Qualitative

Scientific research

Observation protocols and structured interviews

Good for tentatively checking ideas

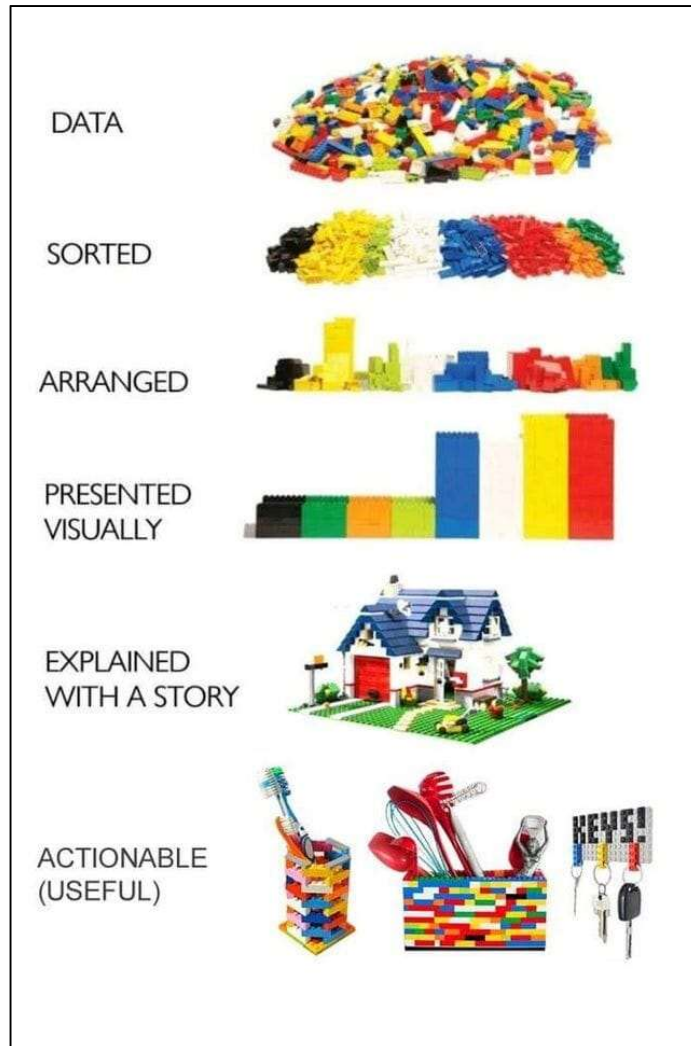
Structured

Highly controlled measures and conditions
(e.g., lab experiment)

Good for testing hypotheses

Quantitative

Design and Engineering



Scientific research

Theory is always involved:

- What are data?
- What are relevant data?
- Sort into which classes?
- Why arranged like that?
- What do you represent?
- Representation = reality?
- Stories are biased framings (security or violation of privacy, different perspectives on the data)
- What is useful depends on culture and context
- 'Use' depends on goals you have or not (e.g., is spyware useful? I have no use for it)

Design and Engineering

Keep in mind that adding more structure to data is adding more conceptual knowledge, which is 'theory'

Whether intuition, tentative knowledge, layman's theory, common sense, or best practice; all of that is based on implicit and thus theoretical assumptions

So if you want to structure data, better read up on the scientific literature →

Qualitative

Scientific research

Observation protocols and structured interviews

Good for tentatively checking ideas

Structured

Highly controlled measures and conditions
(e.g., lab experiment)

Good for testing hypotheses

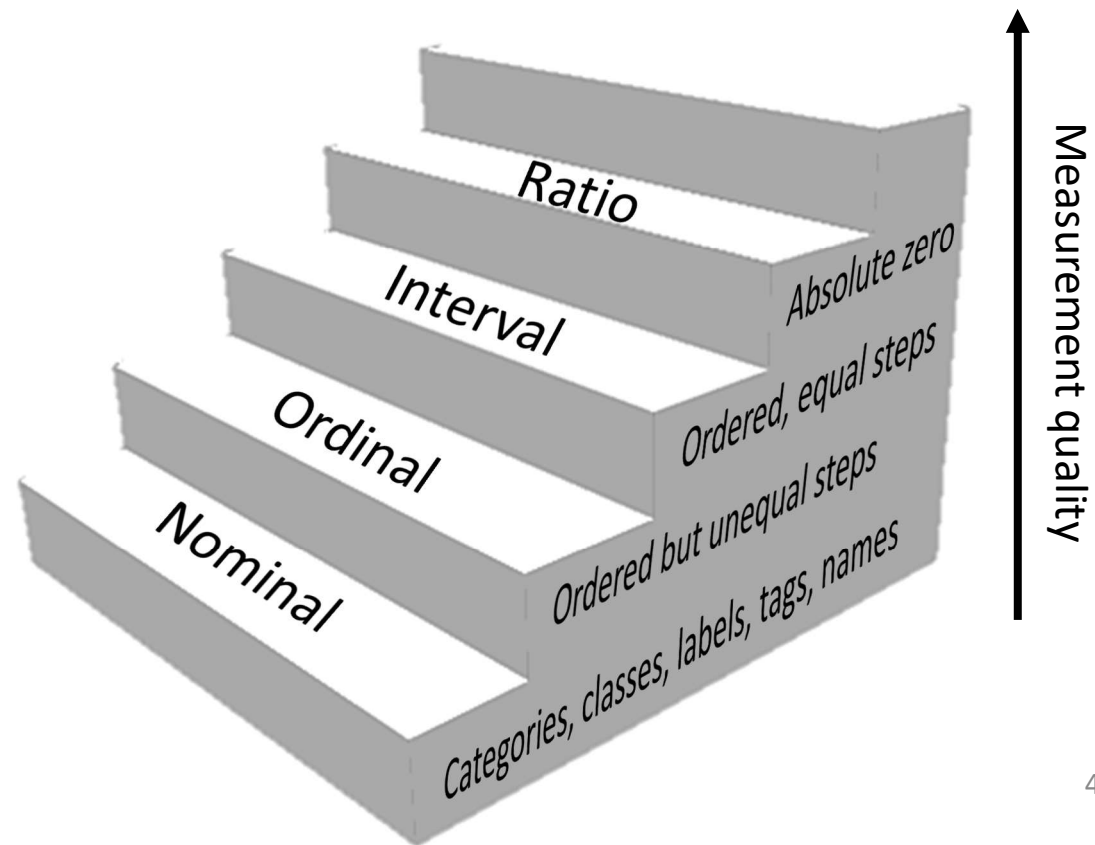
Quantitative

Measures

There are four major scales of measurement:

- Nominal attributes are named, weakest form of measurement
- Ordinal distances between steps bear no meaning
- Interval distances are meaningful
- Ratio absolute 0, strongest measurement

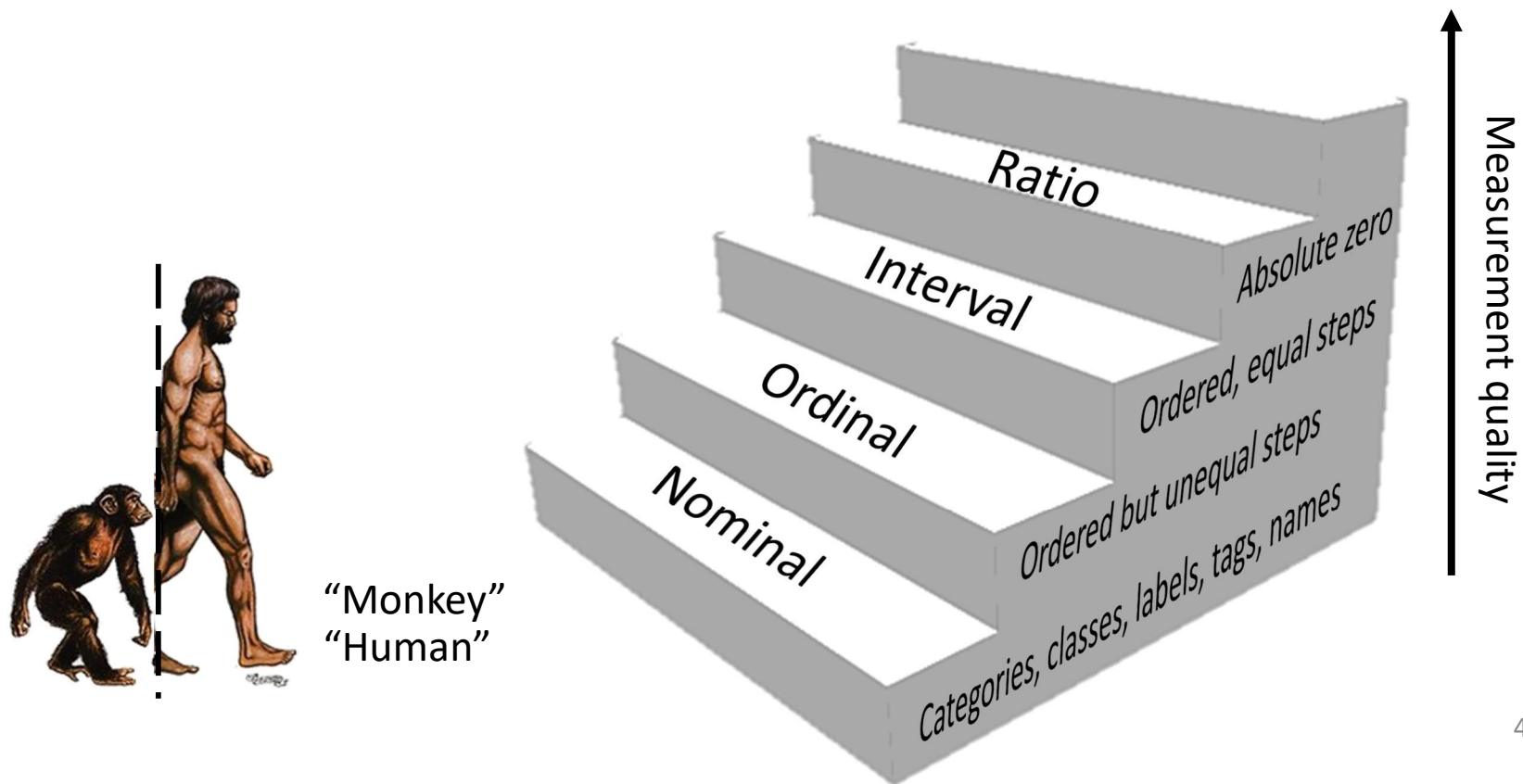
Not all scales are comparable



Measures

There are four major scales of measurement:

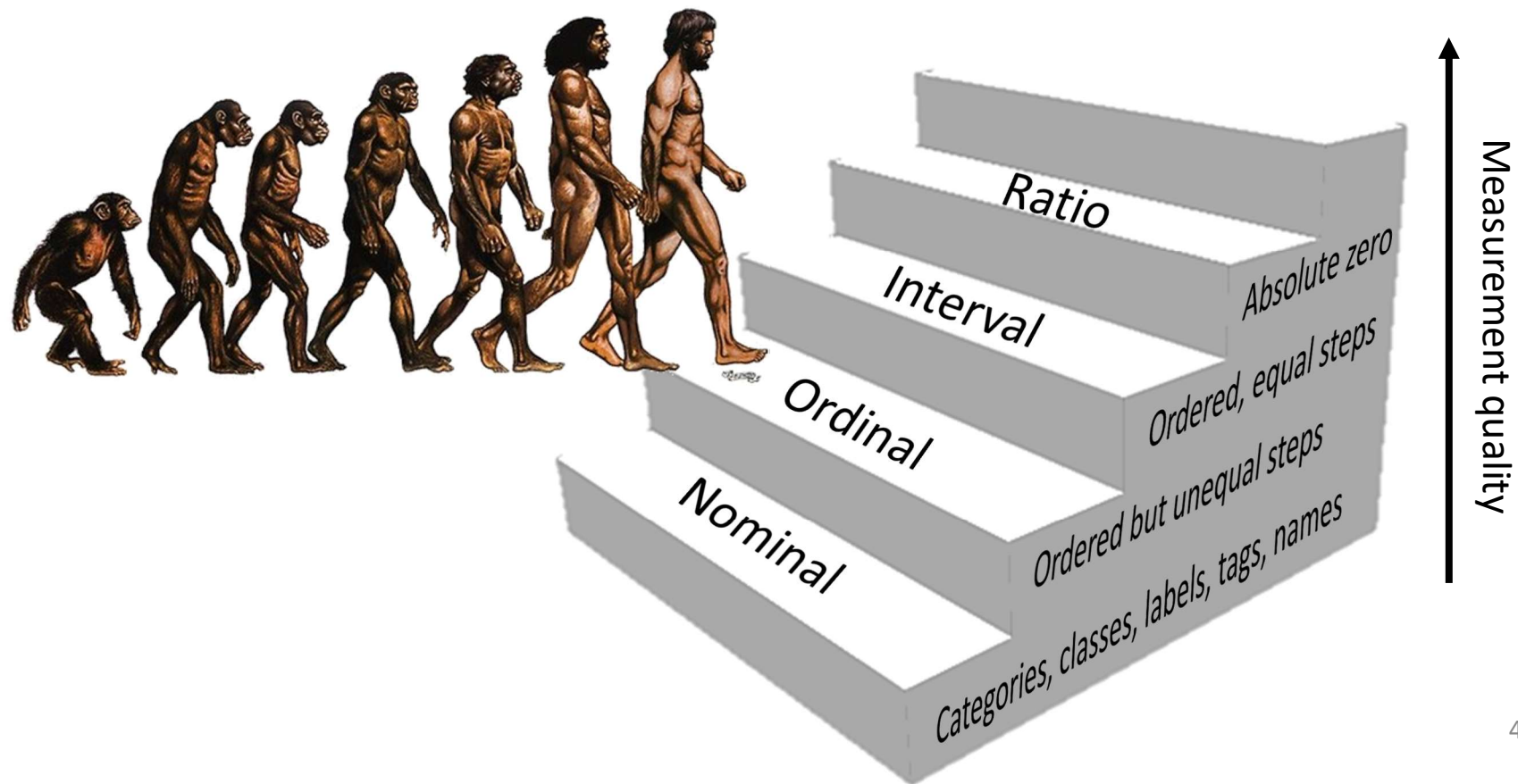
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Measures

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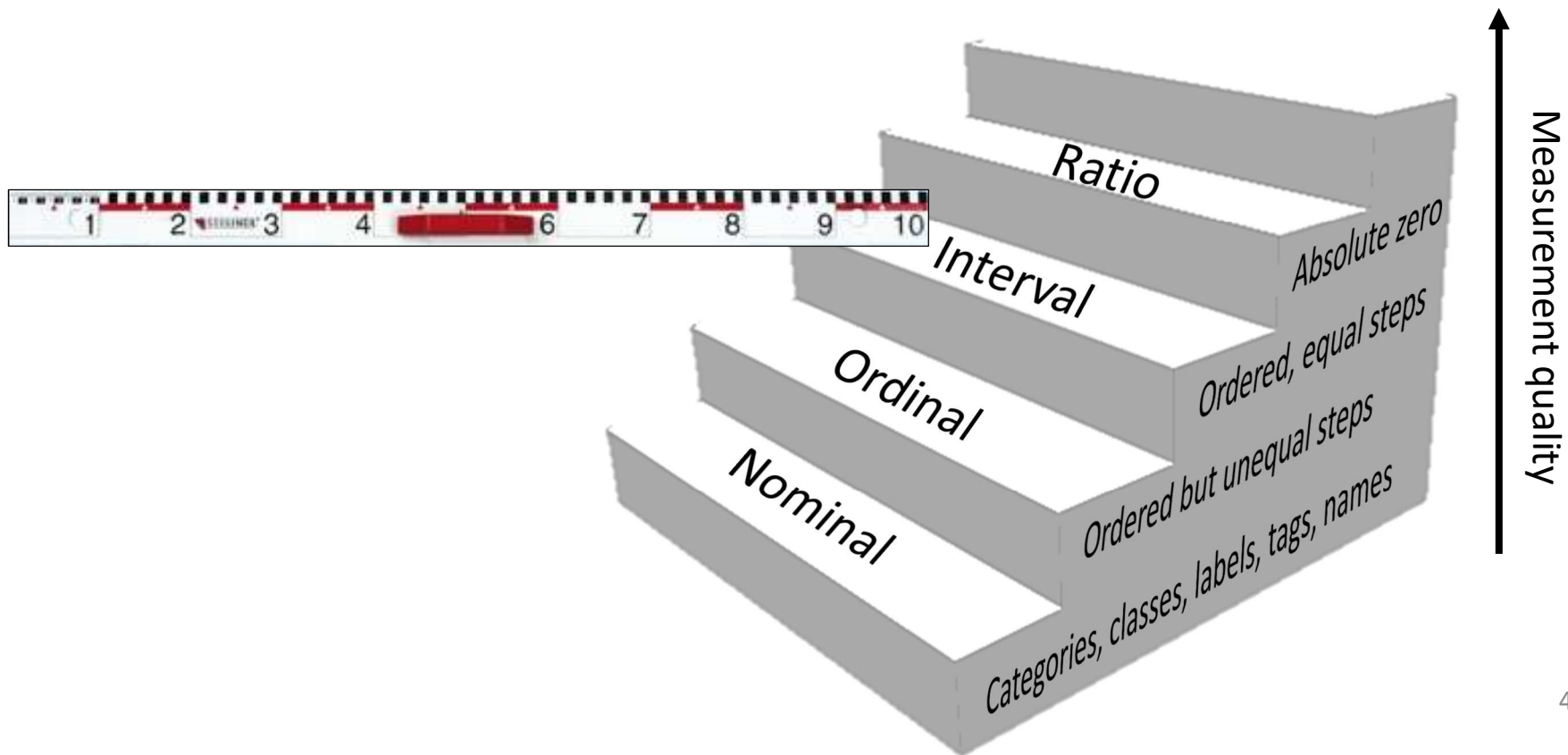
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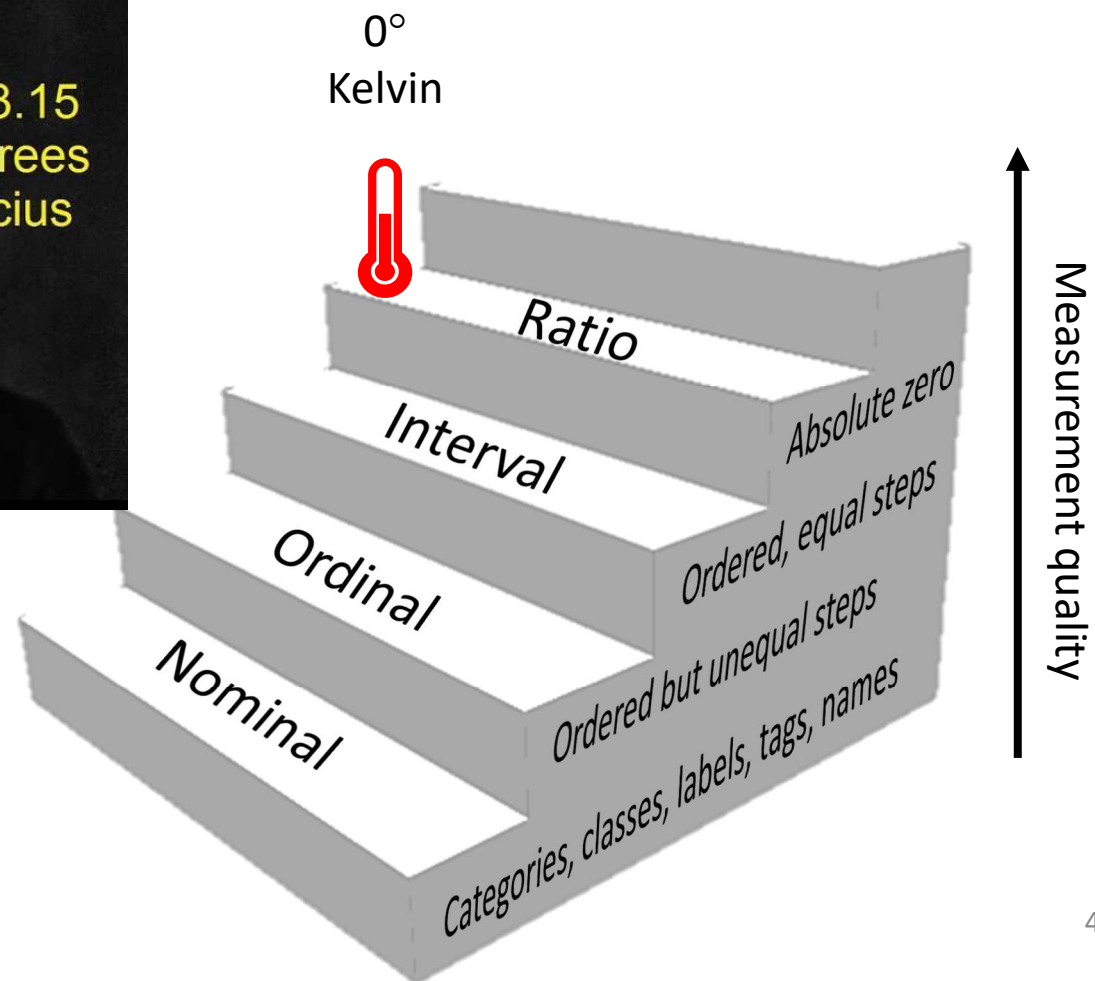
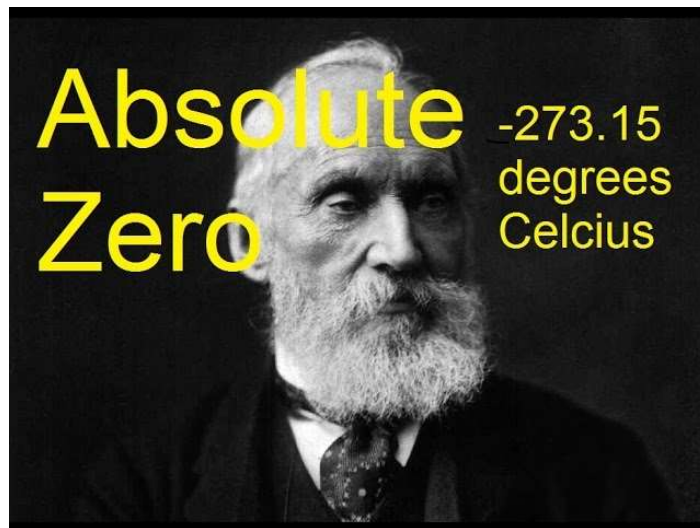
- Interval distances are meaningful



Measures

There are four major scales of measurement:

- Ratio absolute 0, strongest measurement



Measures

☐ Nominal

- Attributes are named, weakest form of measurement
- Categories, classes, labels, tags, names
- Classifies data points into named or numbered but unordered categories (e.g., user groups, gender, religion, education, region)
- Allowable manipulations: number of data points in a category
- Statistics: number of cases in each category, most frequent category

What OS do you use most often on a daily basis?

- ☐ Mac
- ☐ PC
- ☐ Linux
- ☐ Android
- ☐ I don't have a computer

India ▼

Select country

Japan

South Korea

China

India

Mexico

Nigeria

Brazil

United States

Bangladesh

Iran

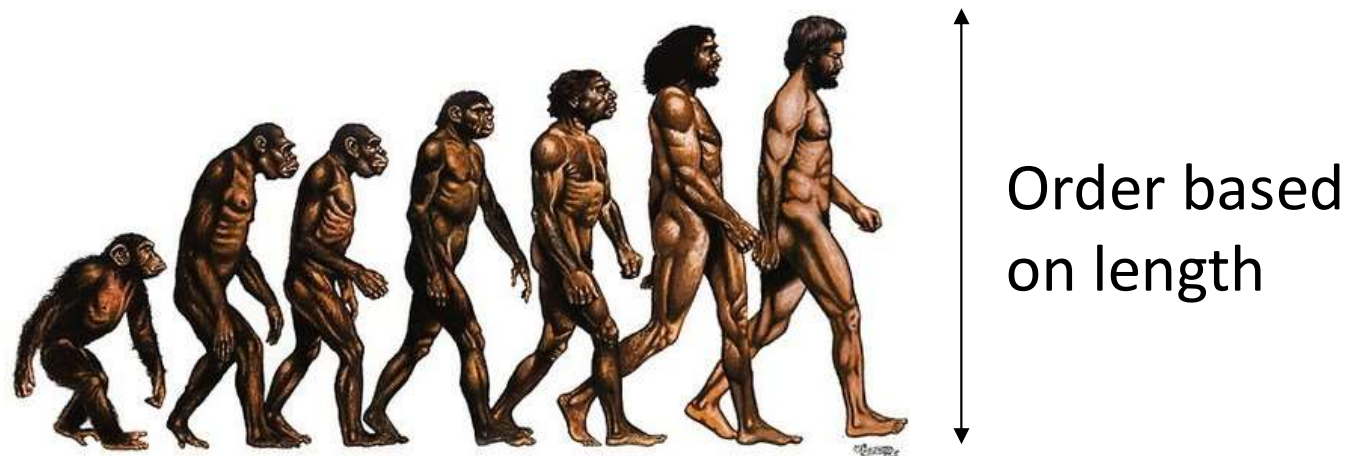


Measures

□ Ordinal scale

- Classifies data points into ordered categories, but differences between each category are unknown
 - Ordered but unequal steps
 - Distances between steps bear no meaning
- Examples: SFQ scores, user preferences
- Allowable manipulations: Counting number of data points in a given category, merging adjacent classes (e.g., number of users who are satisfied with your design)
- Statistics: median, mode, percentiles (30% of users were satisfied)

Type of conclusion: “People were more likely to be satisfied with browsing the Internet on an iPhone than on an Android machine” but you cannot say how much more likely

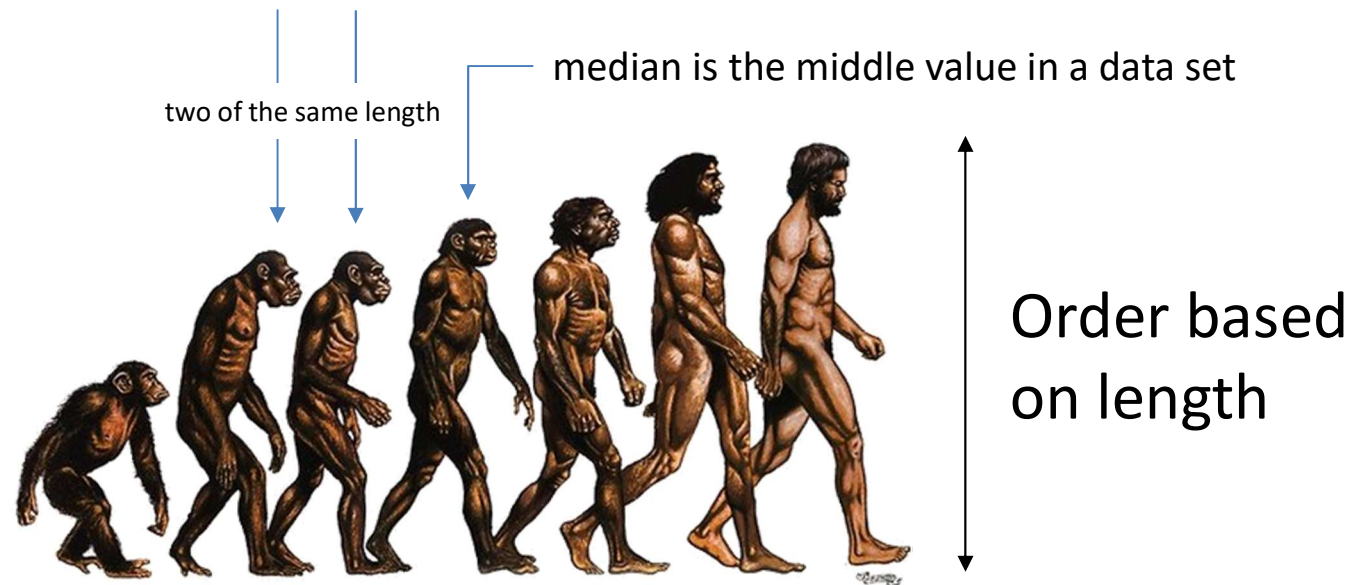


Measures

□ Ordinal scale

- Statistics: median, mode, percentiles (30% of users were satisfied)

mode is the number that occurs most often

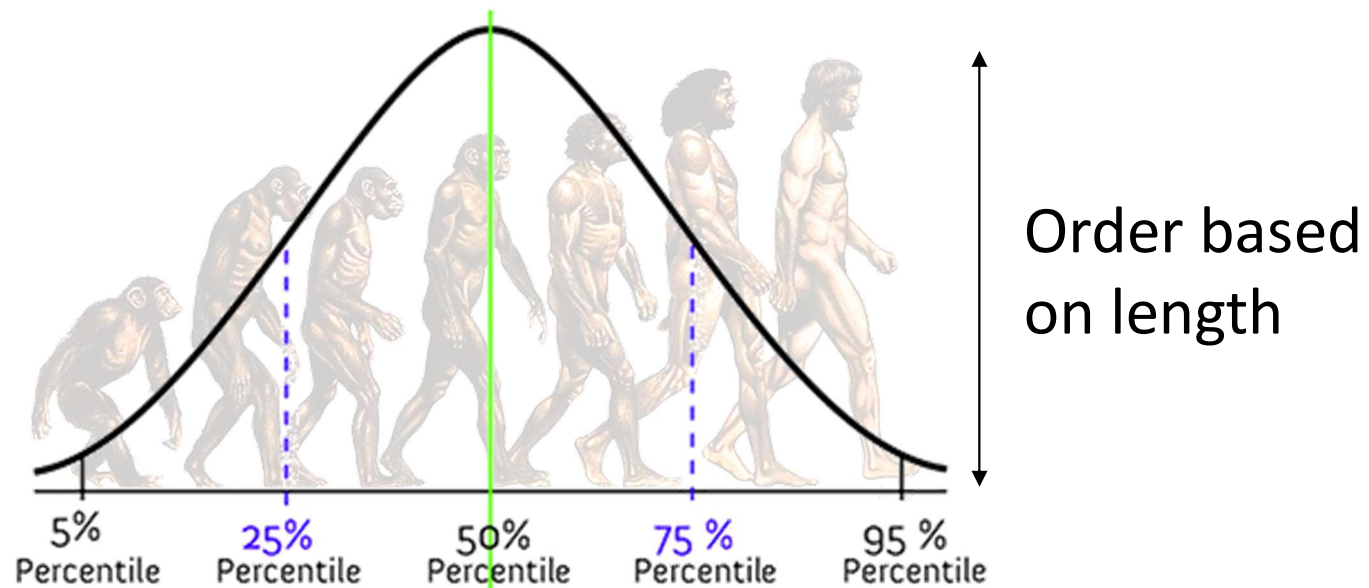


Measures

□ Ordinal scale

- Statistics: median, mode, percentiles (30% of users were satisfied)

percentiles describe how scores in a data set compare



Measures

□ Interval scale

- Similar to Ratio scale, but the “zero” point is arbitrary
- Ordered but with equal steps (distances are meaningful)
- Allowable manipulations: add, subtract
 - Sometimes, ratios (multiply, divide) may not be meaningful
- Allowable statistics: mean, range, variance, standard deviation

Example: A credit score. Three-digit number between 300 and 850 (where's the zero?), indicating credit risk, or the likelihood someone pays their bills



May still carry some ordinal aspects

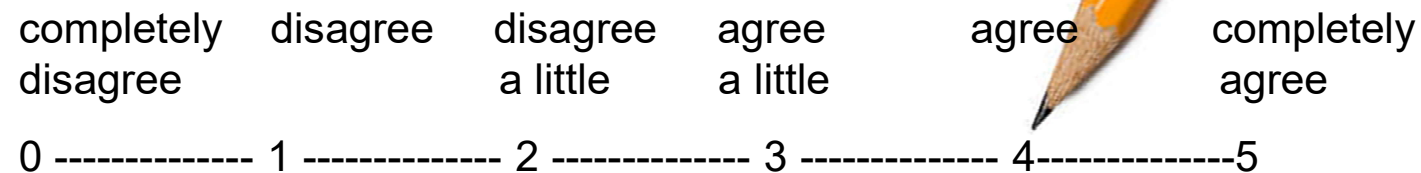
Measures

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Strictly speaking, rating-scale labels are ordinal but statistically can be treated at interval level



Measures

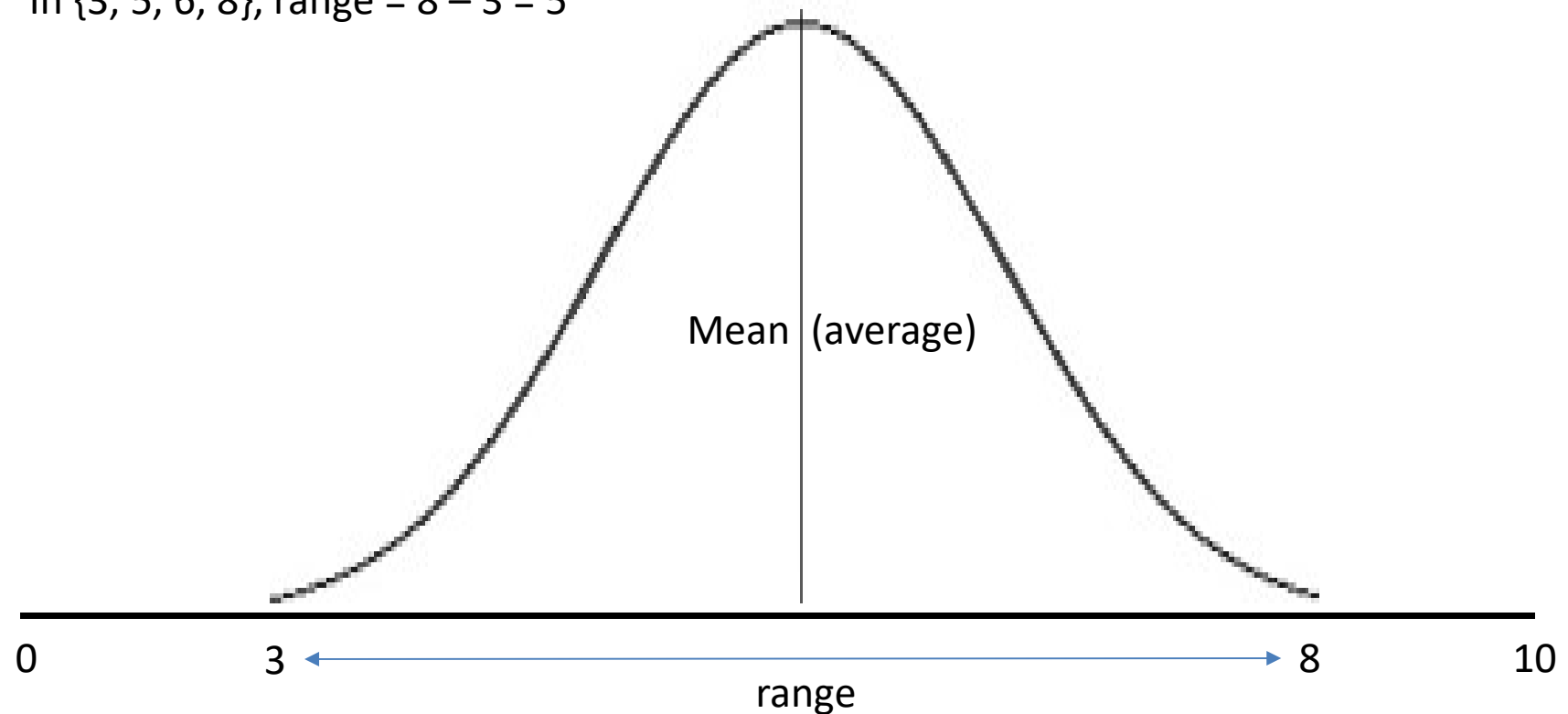
□ Interval scale

- Allowable statistics: mean, range, variance, standard deviation

The mean (average score): The sum of all values divided by the number of values

Range: difference between the highest and lowest scores

In $\{3, 5, 6, 8\}$, range = $8 - 3 = 5$



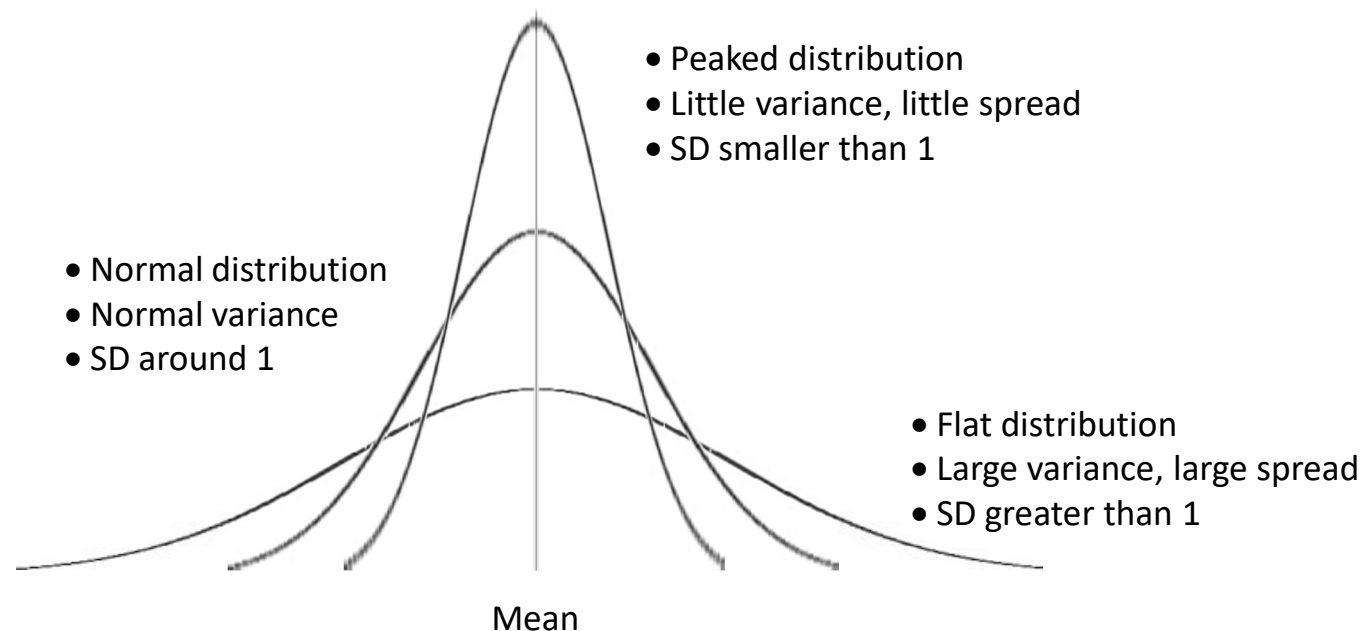
Measures

□ Interval scale

- Allowable statistics: mean, range, variance, standard deviation

Variance: How much each score is different from the mean and thus from every other value in the data set

Standard Deviation (SD): Tells the distribution of a set of data points in relation to the mean value, expressed as 'the square root of the variance'



Measures

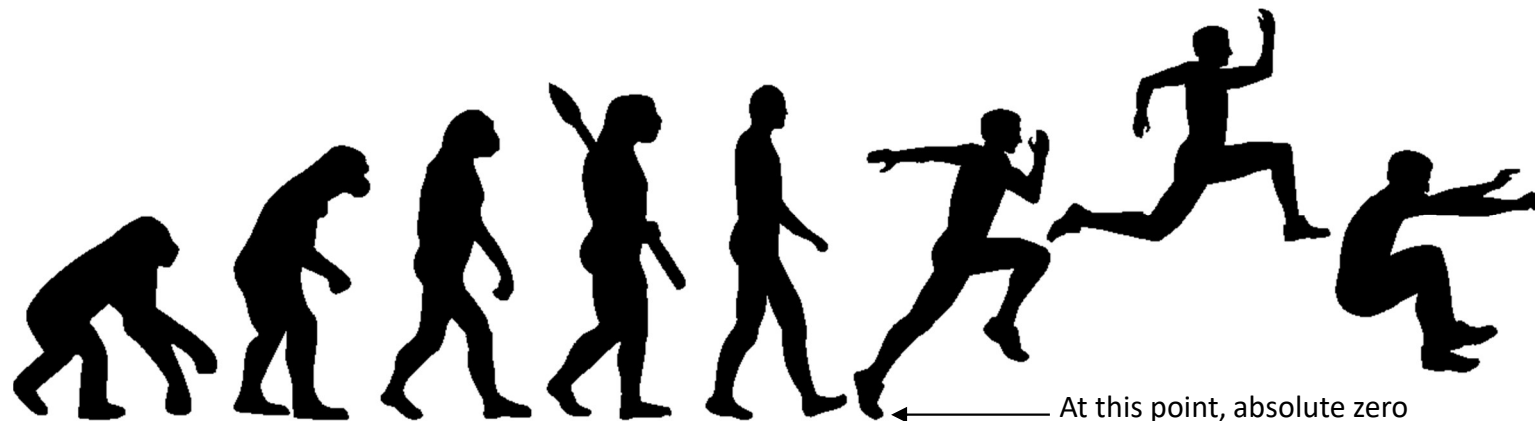
☐ Ratio scale

- Absolute non-arbitrary zero, strongest measurement
 - Classifies data points into ordered categories
 - Equal differences between categories / steps
 - Allowable manipulations: add, subtract, multiply, divide
 - Allowable statistics: mean, range, variance, standard deviation
- Examples: time taken (stopwatch), weight, temperature (in Kelvin)



Type of conclusions: “People use as much or even more time to enter in text messages on an iPhone keyboard than when they use a Samsung keyboard”

“On average, people make as many or even more clicks on sports websites as compared to insurance websites”



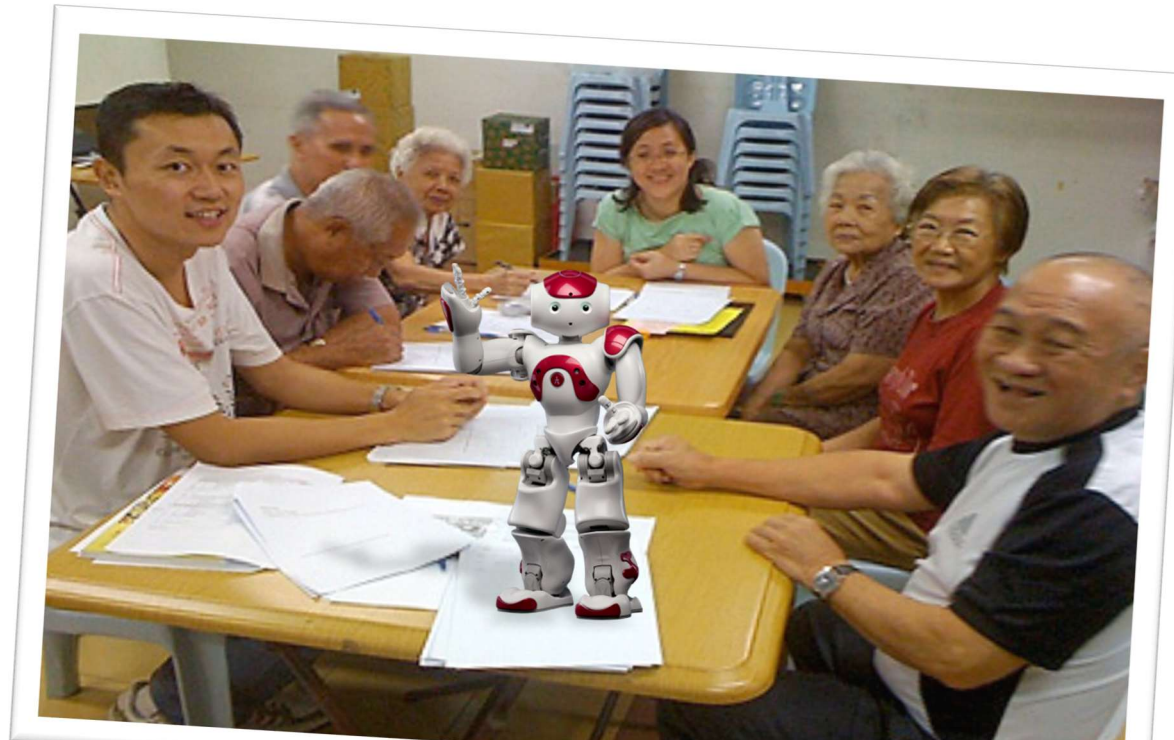
Quantification of qualitative data

- ☐ Use of coding schemes and calculating inter-rater reliabilities

Name		Andrés, Pardo, Díaz, and Angulo (2015), p. 402					
Photo	APPEARANCE						
	Look	Android ☐	Mechanoid ☐	Humanoid ☐	Pet ☐	Toy ☐	
	Age	Adult like ☐	Baby like ☐				
Natural cues  Verbal					Current	Potential	
	Conversational (autonomous dialogue)	Spoken x spoken					
		Spoken x text					
		Text x Spoken					
		Text x text					
	Bidirectional no conversational	Spoken x spoken					
		Spoken x text					
		Text x Spoken					

Quantification of qualitative data

- ❑ Use of coding schemes and calculating inter-rater reliabilities
 - For example, focus-group data with stories of personal experiences
 - Invite at least 3 evaluators (called 'raters') selected for certain demographics (e.g., same education, same profession)
 - Train them for scoring certain aspects (e.g., level of user engagement, opinions, attitude) in the footage / speech / texts
 - You can use a tool for that, like Atlas.ti



Quantification of qualitative data

❑ Use of coding schemes and calculating inter-rater reliabilities

• Calculate inter-rater reliabilities

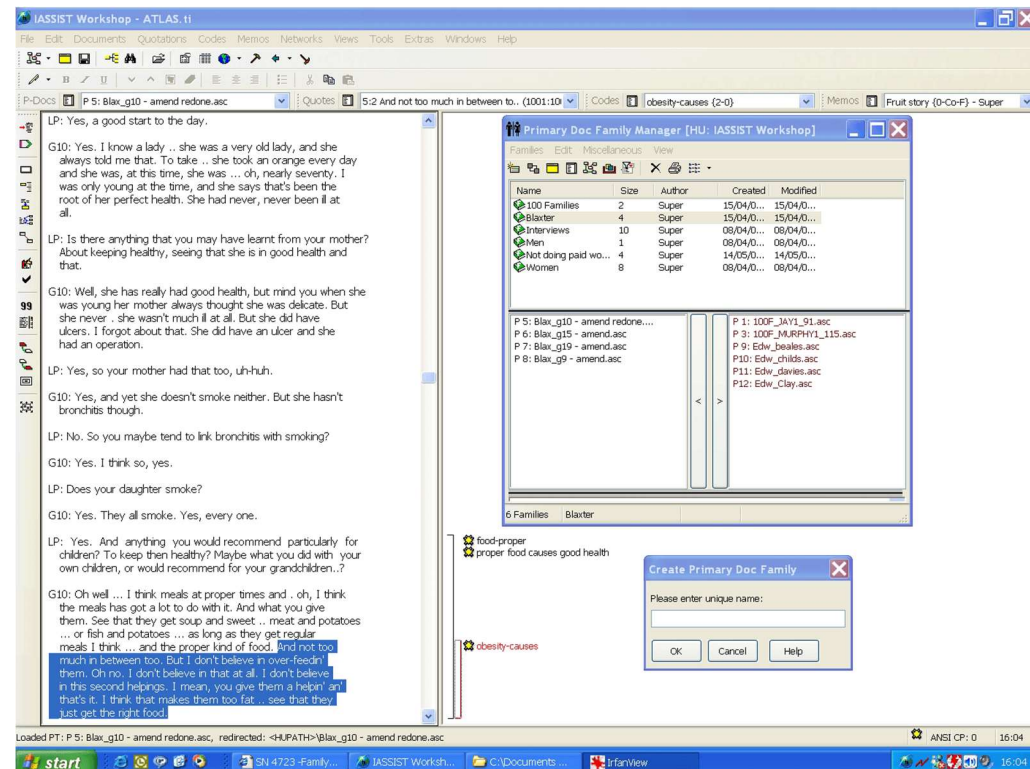
- Percentage agreement (but no control of coincidental agreement)
- Cohen's Kappa
- Krippendorff's alpha



• When reliably consistent, collect raters' scores on the actual data (not the test set)

• Apply appropriate statistics

- Calculate averages
- Correlations
- Regression (multiple)
- *t*-tests for comparing means (e.g., user experience with MacOS or PC)
- Analysis of Variance, etc.



Quantification of qualitative data

❑ Machine analysis

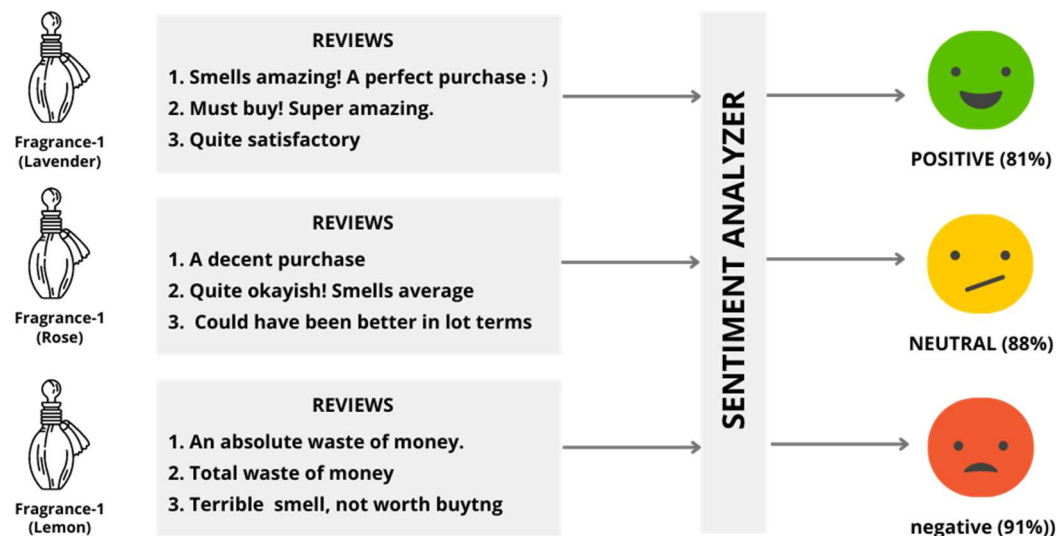
- For instance, sentiment analysis or opinion mining (positive, negative, neutral)

- happy, angry, sad
- urgent, not urgent
- interested, not interested

Can be done in a 'graded' manner: very positive, a little positive, positive (like a rating scale)

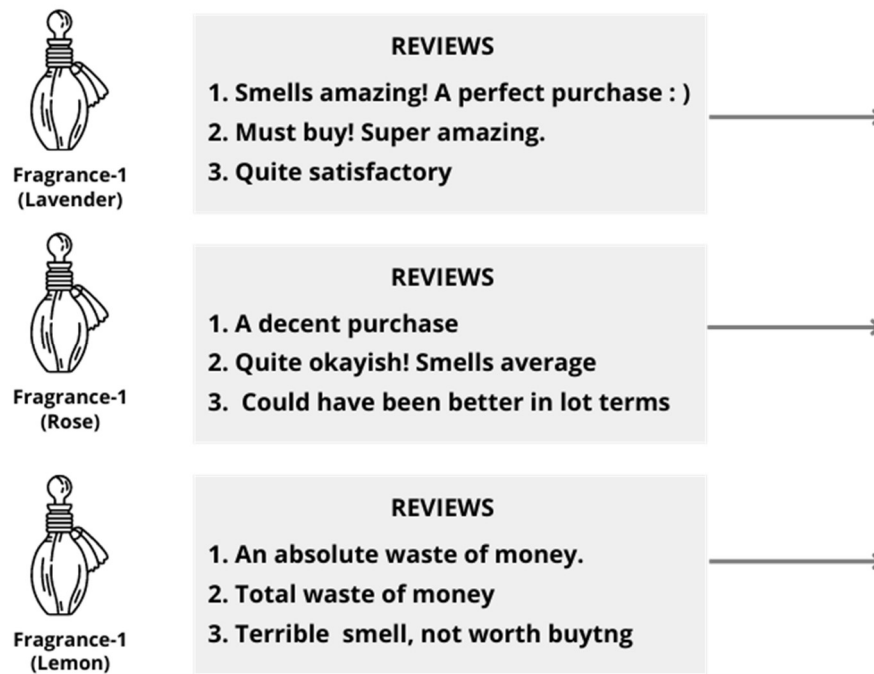
- Again, calculate inter-rater reliability with machine output as one of the 'raters'

Decide to what degree machine analysis is trustworthy



Quantification of qualitative data

- ❑ Qualitative results as input to structured questionnaires



Calculate averages over the scores for indicative and counter-indicative items and you will find how 'neutral' they are

Indicative items

Lavender smells good

completely disagree disagree disagree a little agree a little agree completely agree
0 ----- 1 ----- 2 ----- 3 ----- 4 ----- 5

Lavender smells nice

completely disagree disagree disagree a little agree a little agree completely agree
0 ----- 1 ----- 2 ----- 3 ----- 4 ----- 5

The fragrance of lavender is appealing

completely disagree disagree disagree a little agree a little agree completely agree
0 ----- 1 ----- 2 ----- 3 ----- 4 ----- 5

Counter-indicative items

Lavender smells bad

completely disagree disagree disagree a little agree a little agree completely agree
0 ----- 1 ----- 2 ----- 3 ----- 4 ----- 5

Lavender smells awful

completely disagree disagree disagree a little agree a little agree completely agree
0 ----- 1 ----- 2 ----- 3 ----- 4 ----- 5

The fragrance of lavender is unappealing

completely disagree disagree disagree a little agree a little agree completely agree
0 ----- 1 ----- 2 ----- 3 ----- 4 ----- 5

Measures, issues

❑ Unnecessary downgrading of measurement quality

- High resolution can be degraded to low resolution with certainty
- Low resolution can never be upgraded to higher resolution with certainty
- Missing data, data you did not measure, can be estimated, they cannot be known

Ratio scale

What is your age? _____ years

Ordinal scale

What is your age?

- ☐ Under 16 years
- ☐ 16-25 years
- ☐ 26-49 years
- ☐ 50-65 years
- ☐ Over 65 years

down-gradable

not upgradable

Measures, issues

❑ Unnecessary downgrading of measurement quality

- High resolution can be degraded to low resolution with certainty
- Low resolution can never be restored to high resolution with certainty
- Missing data, data you did not measure, can be estimated, they cannot be known

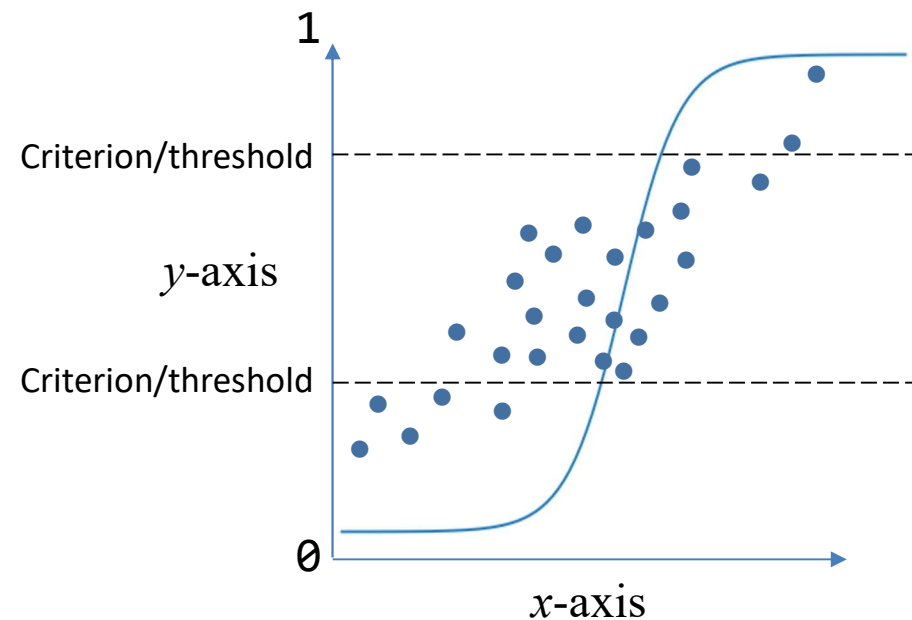
 Like ☒ Totally agree ☐ Agree ☐ Agree a little ☐ Disagree a little ☐ Disagree ☐ Totally disagree OK	Nominal	 Dislike ☒ Totally agree ☐ Agree ☐ Agree a little ☐ Disagree a little ☐ Disagree ☐ Totally disagree OK
now Interval		

Measures, issues

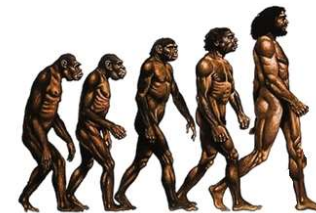
❑ Classifiers in Neural Networks may reduce measurement quality!

- Classifiers split a data set in two or more groups.
By splitting, intermediate values are lost

→ Reduction from interval scale to nominal...



More into the in-class direction



...nuance is lost

More into the out-of-class direction



Measures, issues

- ❑ Classifiers in Neural Networks may reduce measurement quality!

Which may lead to
painful mistakes



Measures

☐ In sum

Scale	Nominal	Ordinal	Interval	Ratio
<i>Frequency distribution</i>	Yes	Yes	Yes	Yes
<i>Median, mode, percentiles</i>	No	Yes	Yes	Yes
<i>Add, subtract</i>	No	No	Yes	Yes
<i>Mean, range, variance, standard deviation</i>	No	No	Yes	Yes
<i>Ratios, coefficient of variation</i>	No	No	No	Yes

Usability testing



UC Berkeley Professor smashes his laptop in front of students to make a point about expectations
<https://laughingsquid.com/wp-content/uploads/2018/01/eric-paulos-smashing-computer.gif>

Usability testing: formative / summative

All formal/informal types of tests can be used in a formative or summative manner

- ❑ Formative means 'during design.' Information helps shape the next iteration
- ❑ Summative means 'after design.' Information is for final evaluation after rollout

- Observations
 - Heuristic testing
 - Cognitive walkthrough
 - Thought listing
 - Focus groups
 - Interviews
 - Lab and field experiments
- *concurrent*
 - *retrospective*
 - *constructive*



Usability testing

Different types of usability testing (Jaspers, 2009), based on input from experts and/or user input:

☐ Observations

- Heuristic testing > experts test rules of thumb
- Cognitive walkthrough > expert acts as typical user
- Thought listing > concurrent, retrospective, constructive

☐ Focus groups

- Moderator-guided discussion

☐ Interviews

- Unstructured
- Structured (semi)

☐ Lab and field experiments.

Work if there is a specific thing that we want to assess (e.g., testing a hypothesis), answering specific questions about targeted parts of a theory or system

Usability testing

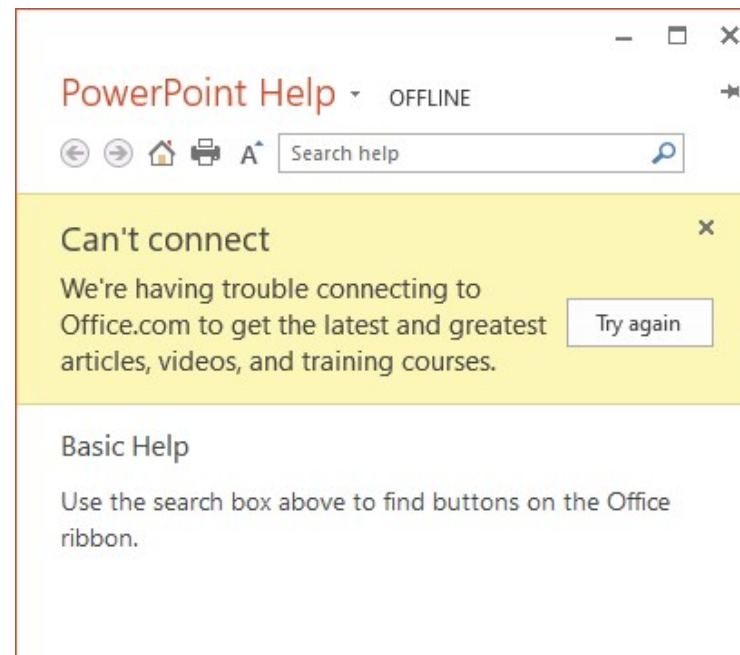
☐ Observations

- Heuristic testing

Evaluators, often experts, examine the interface and judge its compliance with recognized usability principles

Experts score a number of variables according to design guidelines, for example:

- Understandable language, dialogs
- Suited for target group
- Consistency
- Clear manual and help function



Usability testing

- Heuristic testing

The Nielsen-Molich heuristics state that a system should:

1. Keep users informed about its status appropriately and promptly
2. Show information in ways users understand from how the real world operates, and in the users' language
3. Offer users control and let them undo errors easily
4. Be consistent so users aren't confused over what different words, icons, mean
5. Prevent errors - a system should either avoid conditions where errors arise or warn users before they take risky actions ("Are you sure you want to do this?")
6. Have visible information, instructions, etc. to let users recognize options, actions, instead of forcing them to rely on memory
7. Be flexible so experienced users find faster ways to attain goals
8. Have no clutter, containing only relevant information for current tasks
9. Provide plain-language help regarding errors and solutions
10. List concise steps in lean, searchable documentation for overcoming problems

How to conduct heuristic evaluation

1. Know what to test and how - Whether it's the entire product or one procedure, clearly define the parameters of what to test and the objective
2. Know your users and have clear definitions of the target audience's goals, contexts, User personas can help evaluators see things from the users' perspectives
3. Select 3 - 5 evaluators, ensuring their expertise in usability and the relevant industry
4. Define the heuristics (around 5 - 10) - This will depend on the nature of the system / product / design. Consider adopting / adapting the Nielsen-Molich heuristics and / or using / defining others
5. Brief evaluators on what to cover in a selection of tasks, suggesting a scale of severity codes (e.g., critical) to flag issues
6. First walkthrough - Have evaluators use the product freely so they can identify elements to analyze
7. Second walkthrough - Evaluators scrutinize individual elements according to the heuristics. They also examine how these fit into the overall design, clearly recording all issues encountered
8. Debrief evaluators in a session so they can collate results for analysis and suggest fixes

Heuristic evaluation - pros and cons

Pros	Cons
Evaluators can focus directly on specific issues: good, bad points	Evaluators can mark “issues” that aren’t actually usability problems
Evaluators can pinpoint faults with individual elements early on and determine their impact on overall usability	You have to choose exactly the right heuristics to make sure evaluations find <u>all</u> usability issues. Definition issues: what is <u>understandable</u> language?
Compared with lab experiment, you can get feedback without the ethical and practical dimensions and costs – relatively cheap and fast	Number and quality of experts: It can be hard to find evaluators who are experts in a certain industry (e.g., banking) as well as usability
You can combine it with lab experiment	Little structure: As you need several expert evaluators anyway, you may yet want to do lab experiments
With the right heuristics, evaluators can help flag detailed issues and lead the way to optimal solutions	It’s subjective: Findings can lack proof and be biased, and solutions don’t arise automatically

Usability testing

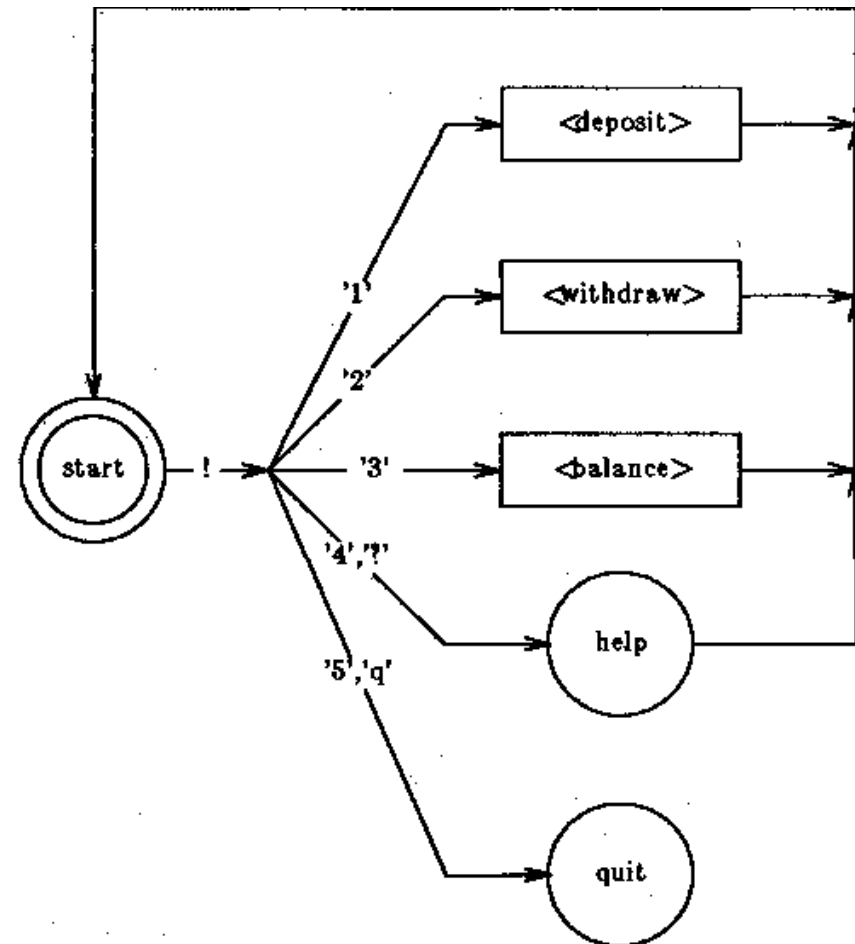
- Cognitive walkthrough

Evaluators are experts and work through a series of tasks and ask a set of questions from the perspective of the user. Expert pretends to be a typical user, solving a specific task. Offers first interaction description with overview of issues:

Example of a task

User login: Open browser >
Navigate to site > Click login button >
Enter username in username field >
Enter password in password field >
Click the login button

Tasks can also be presented as diagrams



Usability testing

- Cognitive walkthrough

Examples of questions to ask yourself:

- Will the user do what you would like?
- Will the user try and achieve the right outcome?
- Does the user see that the right action is possible?
- Will the user notice that the correct action is available to them?
- Does the user associate the action with the goal?
- Will the user associate the correct action with the outcome they expect to achieve?
- Does the user realize that s/he can achieve the goal?
- If the correct action is performed; will the user see that progress is being made towards their intended outcome?

Cognitive walkthrough - pros and cons

Pros	Cons
More structured	Finds task-related issues only
Provides answers to specific questions	Time consuming
	Task description is not easy
	Quality of expert (to simulate a user)

Usability testing

- Thought listing

Potential users (non-experts) perform a task and speak their minds. Analysis protocol by expert, identifying issues. Three types of thought listing:

1. *Concurrent thought listing*

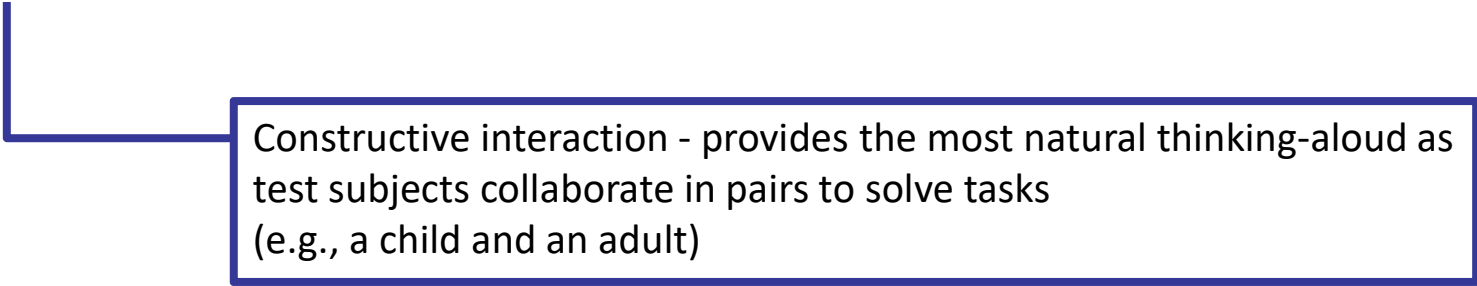
One user at a time, verbalizing while using the application

2. *Retrospective thought listing*

One user at a time, verbalizing thoughts after use

3. *Constructive interaction*

Two users at the same time, conversing about the application



Constructive interaction - provides the most natural thinking-aloud as test subjects collaborate in pairs to solve tasks (e.g., a child and an adult)

Usability testing

- *Constructive interaction*

Provides natural thinking-aloud as test subjects collaborate in pairs to solve tasks

How to conduct constructive interaction (i.e. thinking-aloud protocol)?

Design a set of tasks (like cognitive walkthrough)

Participants are asked to say whatever comes to mind as they complete the task

- what they are looking at
- what they are thinking about
- what they are going to do

Record and observe (without attempting to interpret their actions and words)

Sessions are often audio- and video-recorded so that developers can go back and refer to what participants did and how they reacted

Thought listing - pros and cons

Pros	Cons
Real users, rich data	Time-consuming
Good insight into problems	Selection of users, choice of task
	Are all thoughts verbalized?

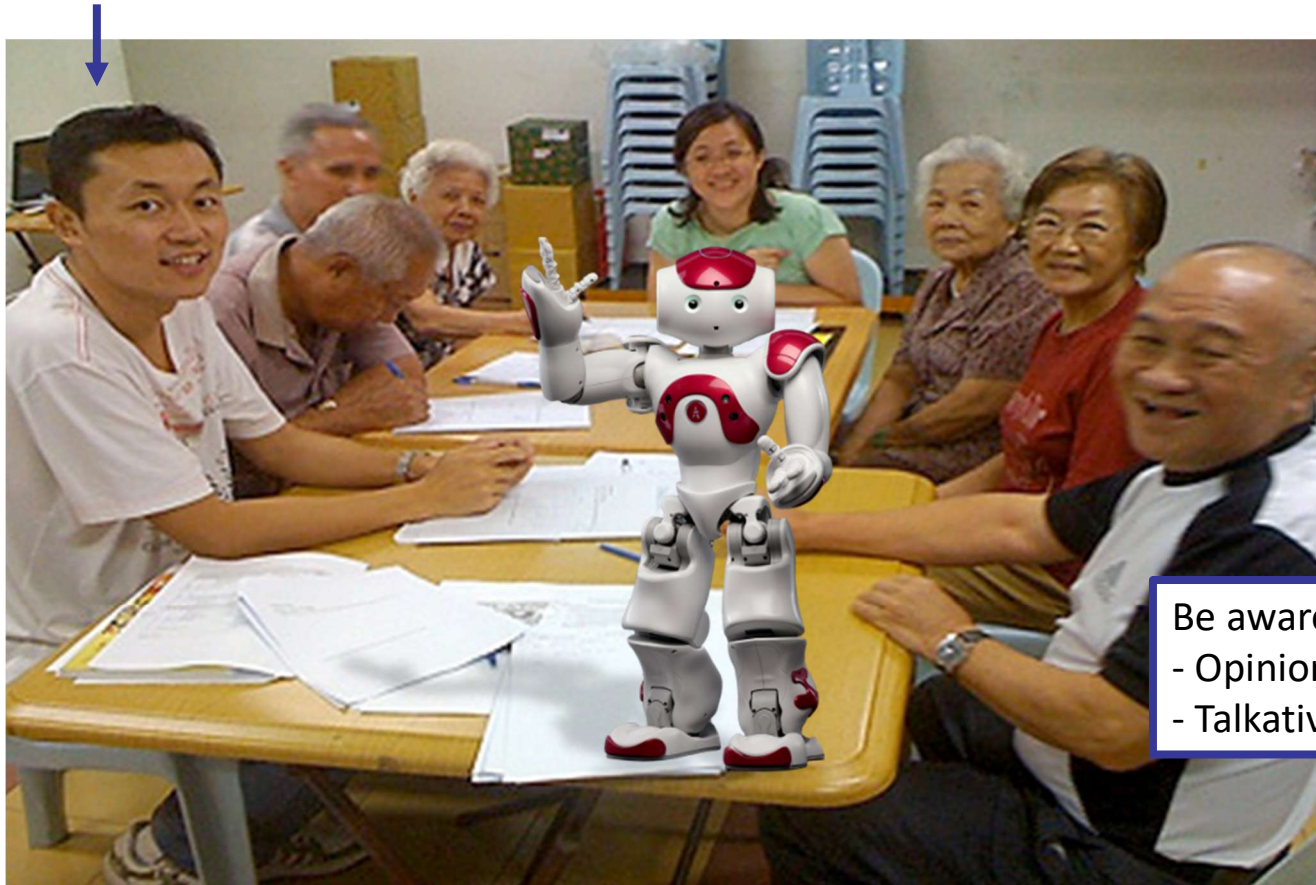
Usability testing

☐ Focus groups

Complements *constructive interaction*

Small groups, predefined demographic traits (e.g., older adults, caretakers)

Moderated, questions directed at particular topic (e.g., ethical use, societal impact)



Be aware of group dynamics:

- Opinion leaders
- Talkative vs shy

Usability testing

☐ Interviews

Oriented at the individual, complement of focus groups

- Unstructured

Open format, like a normal conversation, no questions prepared, but topic defined
Good for sensitive topics (e.g., robots against loneliness and depression)

- Semi-structured

Has an interview protocol, guided conversation, has structure as well as options built in to go into specific details, aspects, issues that were unforeseen

- Structured

Almost like a questionnaire but face-to-face. Strict interview protocol. May be felt as rigid but is good if specific questions / hypotheses should be investigated.
No options to explore unforeseen topics. Risk of missing out on important aspects but renders more 'correct' answers:

“Thirty percent of the interviewees were against robots in care.”

Usability testing

❑ Lab and field experiments

An experiment is designed to test a **hypothesis** about the role of one variable (the **independent variable**) on another (the **dependent variable**)

Works if there is a specific thing that we want to assess, answering specific questions about targeted parts of a theory or system

- Does my new approach lead to faster completion of tasks?
- Is a text interface easier to remember than a GUI?

For example, to test the validity of Fitts' Law

Relates the distance to the target, the size of the target and the pointing system to the time it takes to point to the target

- The distance and size of the target are known
- Users perform a series of pointing tasks (trials) with different distances and sizes
- Different pointing systems are timed (sec) and measured for accuracy (hits vs misses)

Usability testing: formative

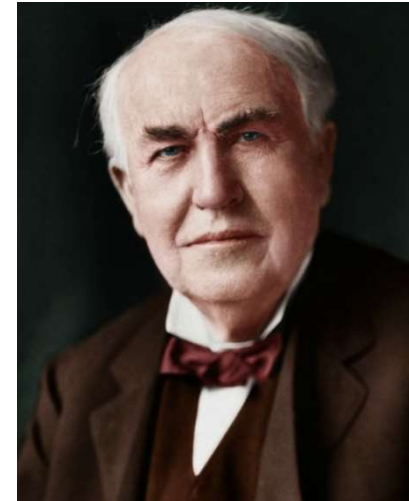
As said, any formal or informal test type can be used for formative or summative testing of your application

❑ Formative (during system development, usually based on a prototype)

- Understandability of application
- Completeness of application
- Quality of the application
- Ethical review (e.g., health apps)
- Liability and law (e.g., self-driving cars)

“I have not failed 1,000 times. I have not failed once. I have succeeded in proving that those 1,000 ways will not work. When I have eliminated the ways that will not work, I will find the way that will work.”

- Fail fast innovation
- Rapid prototyping, user testing
- Stakeholder co-design



Thomas Alva Edison
(1847-1931)

Usability testing: summative

❑ Summative (after system development)

- Process evaluation

Is the application well implemented?

User statistics

Accessibility

Safety, security

code review

click behavior, website traffic, peaks
of website / contact info

cf. Fintech

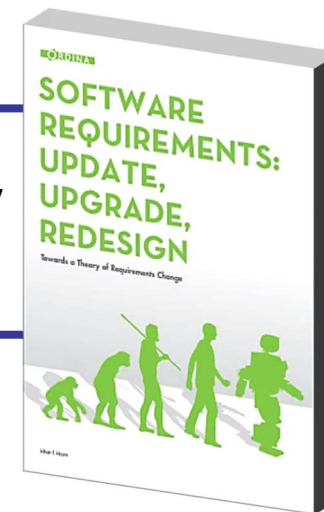
- Impact evaluation (effect on users, other stakeholders)

Environmental and behavioral factors (cf. physical and cognitive ergonomics)

Final effect of the app on someone's:

financial situation, health, work performance, quality of life, etc.

Waterfall models typically restrict themselves to summative testing after rollout to discover that the many change requests make redesign very expensive (money, effort, motivation)



Human-generated summary



User testing is important to inform design decisions: Judgments by experts, lab and field experiments, and field studies with real users avoid market failure (irrespective of proper coding). We do usability tests to validate our desktop assumptions, we don't stay sitting behind our screens

The four quadrants of testing help you determine what kind of test you need or are doing: more abstract or concrete, obtrusive or unobtrusive

Ethnographical observations on the work floor are unobtrusive and concrete, people will not notice you study them but cases are singular and data are anecdotal. Lab experiments are obtrusive and so people may be biased by you observing them but cases are numerous and can be analyzed statistically

Qualitative tests generate non-numerical data (e.g., videos, use stories), which give you specific insights into your particular case. Quantitative tests are numerical (e.g., speed, accuracy) and may be generalized over multiple cases

Human-generated summary



You must always start from formulating a Research Question and derive Hypotheses from it

The Null predicts there is no noticeable difference between two sets of data (e.g., Mac vs PC use): Effects are statistically not significant. The Alternative does predict a difference: The difference is significant beyond reasonable doubt

Start testing while prototyping ('formative'). Don't wait until system roll out ('summative')

Tests with high applicability are:

- Heuristic testing (experts test rules of thumb)
- Cognitive walkthrough (expert acts as typical user)
- Thought listing (concurrent, retrospective, constructive)

But focus groups, un/semi/structured interviews, lab and field experiments can bring valuable additional results to inform your designs

Exam Sample Item

Category “Knowledge”

Jaspers (2009) states that during formative testing of an application, heuristic testing may be an option. In contrast to lab experiments, defining aspects of heuristic testing are:

- A It is a cheap and fast way to let naïve users offer rules of thumb that guide the test
- B It is a slow but thorough way to let experts offer rules of thumb that guide the test
- C It is a slow but thorough way to let naïve users offer rules of thumb that guide the test
- D It is a cheap and fast way to let experts offer rules of thumb that guide the test

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- ☒ D It is a cheap and fast way to let experts offer rules of thumb that guide the test